

Review article

AI-enabled defect detection in industrial products: A comprehensive survey, key insights and future research challenges

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ARTICLE INFO

Keywords:

Object defects
Defect inspection
Defect localisation
Machine learning
Deep learning

ABSTRACT

Artificial Intelligence (AI) techniques, especially Machine Learning (ML) and Deep Learning (DL), are increasingly used for automated defect inspection in industries like metals, ceramics, glass, and textiles. These methods process high-quality images to detect and localise defects. Inspection approaches range from manual and semi-automated (classical ML) to fully automated (DL), with DL preferred for eliminating manual feature engineering. DL models are typically supervised or unsupervised—supervised models offer high accuracy but require large labelled datasets, while unsupervised models work without labels but often yield lower performance. Emerging alternatives like semi-supervised, weakly supervised, and self-supervised learning show promising results in data-scarce settings. This paper presents a comprehensive taxonomy of DL models for defect classification, detection, and segmentation. For classification, models such as Convolutional Neural Networks (CNNs), transfer learning, and hybrid architectures are discussed. For segmentation tasks, encoder-decoder models, pyramid networks, and attention mechanisms are reviewed. In defect detection, the evolution of YOLO architectures is examined, particularly combinations with pyramid networks and attention modules to enhance feature extraction. Unsupervised approaches, such as Generative Adversarial Networks (GANs) and diffusion models, are explored for their capacity to detect anomalies without extensive labelled datasets. Emerging strategies — including transformer-based models, few-shot learning, large language models (LLMs), and foundation models — are also covered. Despite recent advancements, challenges remain in ensuring model reliability due to overconfidence and poor calibration. We address this by discussing model trustworthiness, human-in-the-loop frameworks, and reliability assessment techniques critical for industrial deployment. This review analyses the current strengths and limitations of existing approaches, identifies ongoing challenges, and highlights future directions in ML and DL-based defect inspection across various materials and defect types.

Contents

1.	Introduction	2
2.	Defects on products	3
2.1.	Defect categories and materials	4
2.2.	Defect characteristics	4
2.3.	Defects inspection	5
2.4.	Defects imaging	5
3.	Classification of defect detection techniques	6
3.1.	Semi-automated defect detection techniques	6
3.2.	DL-based automated defect detection techniques	7
3.3.	Architecture of automated defect detection methods	7
4.	Efficacy of defect detection techniques using semi-automated learning	8
4.1.	Image processing-based semi-automated techniques	8
4.2.	Classical machine learning based techniques	8

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4.2.1.	Supervised learning	8
4.2.2.	Unsupervised learning	8
4.2.3.	Hybrid methods	8
5.	Efficacy of DL-based automated defect detection	9
5.1.	Defect classification methods	10
5.1.1.	Supervised learning (CNN-based methods).....	10
5.1.2.	Supervised learning (Transfer-learning-based models).....	10
5.1.3.	Hybridised models.....	10
5.1.4.	Unsupervised learning	11
5.1.5.	Discussion	12
5.2.	Defect segmentation techniques	12
5.2.1.	Encoder–decoder based supervised segmentation.....	12
5.2.2.	Pyramid network-based supervised segmentation	12
5.2.3.	Attention-based supervised segmentation.....	14
5.2.4.	Defect segmentation by unsupervised learning (GAN-based).....	14
5.2.5.	Discussion	14
5.3.	Defect detection techniques.....	14
5.3.1.	YOLO with pyramid-based networks	14
5.3.2.	YOLO with attention-based models	15
5.4.	Unsupervised, semi-supervised and other learning for defect detection.....	16
5.4.1.	GAN-based models	16
5.4.2.	Diffusion-based models	17
5.4.3.	Semi-supervised learning.....	17
5.4.4.	Self-supervised learning	17
5.4.5.	Weekly-supervised learning	17
5.4.6.	Discussion	18
6.	Other networks.....	18
6.1.	Transformer-based models.....	18
7.	Highlights of the study.....	19
8.	Open challenges and future research directions	21
8.1.	Data pre-processing in industrial settings.....	21
8.2.	Deep learning models: Refining network selection and parameter tuning	21
8.2.1.	RQ1: Addressing the problem of small dataset in dl-based models.....	22
8.2.2.	RQ2: Hyperparameter selection for models optimisation	22
8.3.	Defect detection models: Navigating reliability and explainability	23
8.3.1.	RQ3: Reliability challenges and solutions for DL-based models.....	23
8.3.2.	RQ4: Integrating HITLA in DL-based models refinement	24
8.3.3.	Challenges in training time, inference, accuracy and continuous monitoring	24
8.4.	Future industry applications	24
9.	Conclusion	24
	CReditT authorship contribution statement	25
	Declaration of competing interest.....	25
	Data availability	25
	References.....	25

1. Introduction

During the manufacturing of products in complex industrial processes, defects such as internal holes [1], pits [2], and scratches [3] arise on the surface, edges and corners due to errors in machine design which adversely not only affects the quality of the product but also its performance and value [2,4]. Different types of defects are visible on different products (such as scratch on steel [5], blob on textile [6], crease on carbide [7] and chipping on glass [8], etc.). In addition, defects on products not only cause waste of resources, decrease revenue, and diminish customer satisfaction but also add additional cost for identification of defects [9].

Often new and used products are sold on the Internet (e.g., eBay, Amazon, Gumtree and Facebook marketplace). Buyers must rely on the description of the seller given with the product photos. They may have a lack of confidence to decide whether to buy a product of interest. An independent assessment system can be developed, and the inspection report can be attached to the product advertisement to increase buyer's confidence and satisfaction. So, defect detection and classification are an important topic that is treated as a specific problem to the industry at the product manufacturing and production phases [10]. Manual defect identification requires experts to inspect and handle products

physically. Thus, this is not only expensive but also makes the inspection result subjective. Moreover, this poses products to be more defective due to physical handling. For example, Mital et al. [11] uses a physical inspection method for product parts in the industrial sector. It works excellently in some cases but has performance consequences because of poor detection and lower real-time performance [9]. Thus, this method does not satisfy the requirement of quality inspection of modern industrial production [12].

Therefore, a more reliable and efficient methodology should be introduced. Automated defect detection methodology has more advantages over manual identification. It saves time, decreases the overall production cost, eliminates physical product handling, increases product quality, and works in the long run with high efficiency. For example, machine vision techniques can take images of products and allow computers to inspect the images. The use of computers for analysing images and the application of AI for decision-making make the process more automated and reliable [13,14]. Recently, automated methods have been investigated on various products, e.g., pharmaceutical [15], steel [5] and textile [6]. They mainly use computer vision and image processing techniques in surface defect detection [16–18]. Thus, modern manufacturing and industrial processes have been integrated with image processing, computer vision and AI-based systems [12,19, 20] to improve the accuracy of detection and to reduce manpower requirements.

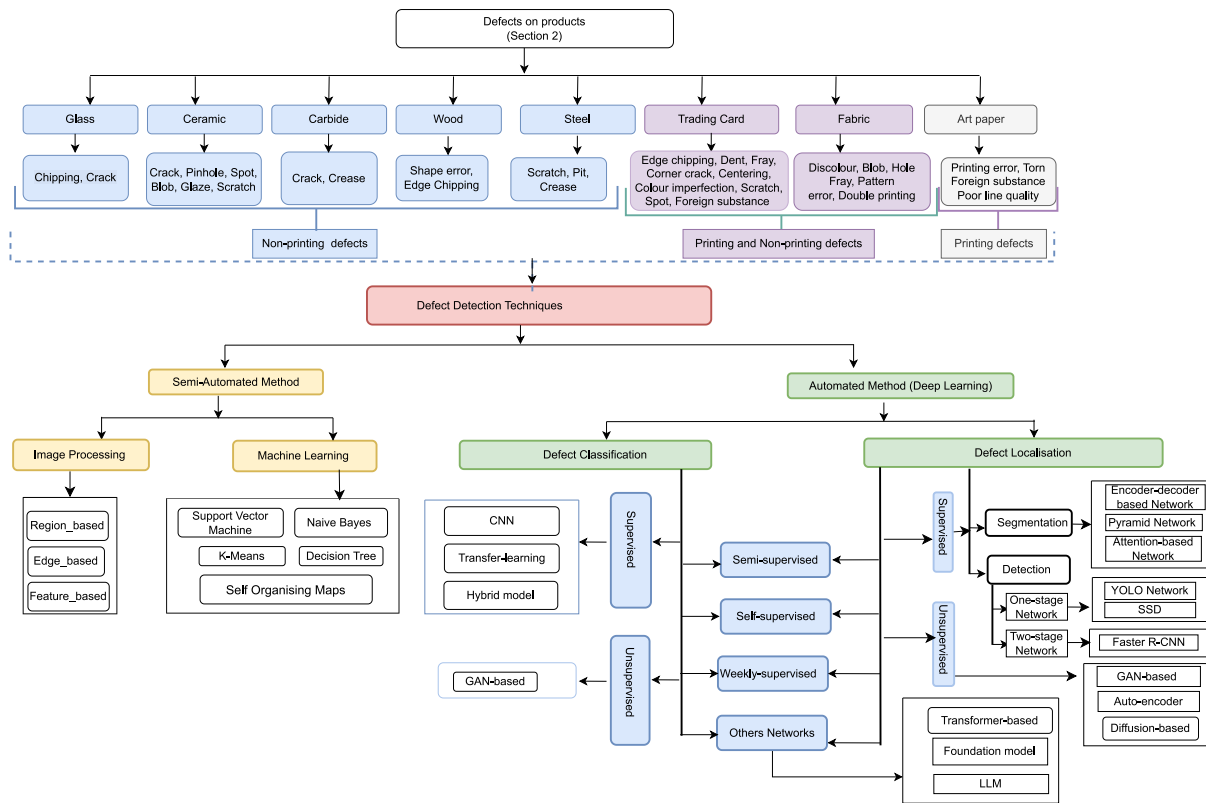


Fig. 1. A taxonomy of product defects and their detection techniques.

Semi-automated defect detection methods, based on classical image processing and machine learning (ML) techniques, rely heavily on manual parameter tuning and feature engineering. Although the application of classical ML techniques (e.g., Support Vector Machine) on the features (e.g., mean, variance, curvature) extracted using image processing techniques (e.g., texture analysis) improves the defect detection performance, the performance is still under the required satisfaction level. In contrast, deep learning (DL) models [21] eliminate manual feature extraction by learning directly from images, enabling full automation, improved performance, and wider adoption in industrial defect classification and localisation.

Furthermore, supervised DL models require large labelled datasets for high accuracy [22], but data collection is labour-intensive. To mitigate this, researchers use data augmentation [23] and synthesis algorithms to address dataset imbalance [24]. Despite these efforts, labelling remains time-consuming. Segmentation helps locate defects [25], yet pixel-level annotation adds cost. As an alternative, unsupervised DL uses unlabelled data, though image scarcity persists in some domains. Dong et al. [26] shows promising results by combining data augmentation with unsupervised segmentation on fewer unlabelled images.

Existing surveys on defect detection [27–33] primarily focus on specific products and predominantly discuss the application of DL techniques. However, these surveys lack in detail descriptions regarding the quantification of defect size, determining the location of defects, and the utilisation of image-capturing devices based on defect size. However, these aspects are important for developing comprehensive defect detection methodologies and improving the accuracy and efficiency of defect analysis in various industries. This study focuses on the state of development of defect detection on industrial products utilising machine vision. Fig. 1 illustrates the overview of this work such as image processing, ML and DL methods, which are discussed and critically analysed in detail, for defect detection.

The key contributions of this paper are listed as follows:

- This study examines the characteristics of different defect types found on industrial products.
- Provides a comprehensive overview of semi-automated and automated methods that are applied to detect defects on industrial products.
- Categorises DL-based automated models that have been employed for defect detection applications.
- Highlights the challenges and summarises the potential research directions for quality assessment of industrial products.

The aim of this review is to integrate and compile all possible information regarding defect classification and localisation on various objects using ML and DL. To achieve this, articles are searched using key terms or synonyms such as defect detection, fault classification, and crack identification from databases including Scopus, IEEE, Elsevier, Springer and Google Scholar. Initially, articles are searched using the full text of the papers based on specific objects (e.g., steel, wood, glass, fabric etc.) and key terms. Duplicate documents and non English article are eliminated. Following this, some articles are removed after reading the abstracts which is irrelevant to the research. Finally, around 160 articles are selected for our survey. This research aims to assist researchers in finding appropriate articles for defect classification and localisation in industrial applications.

The rest of the paper is organised as follows. Section 2 describes types of defects, Section 3 represents overview of different types of defect detection techniques, Section 4 describes the articles regarding semi-automated methods while Section 5 describes the articles about automated methods and other networks covers in Section 6. Section 7 provides a statistical analysis of the literature while Section 8 provides challenges and future research directions, Finally, Section 9 concludes the paper.

2. Defects on products

At present, AI-based methods such as ML and DL have grown quickly in the context of data analysis and inspecting product quality

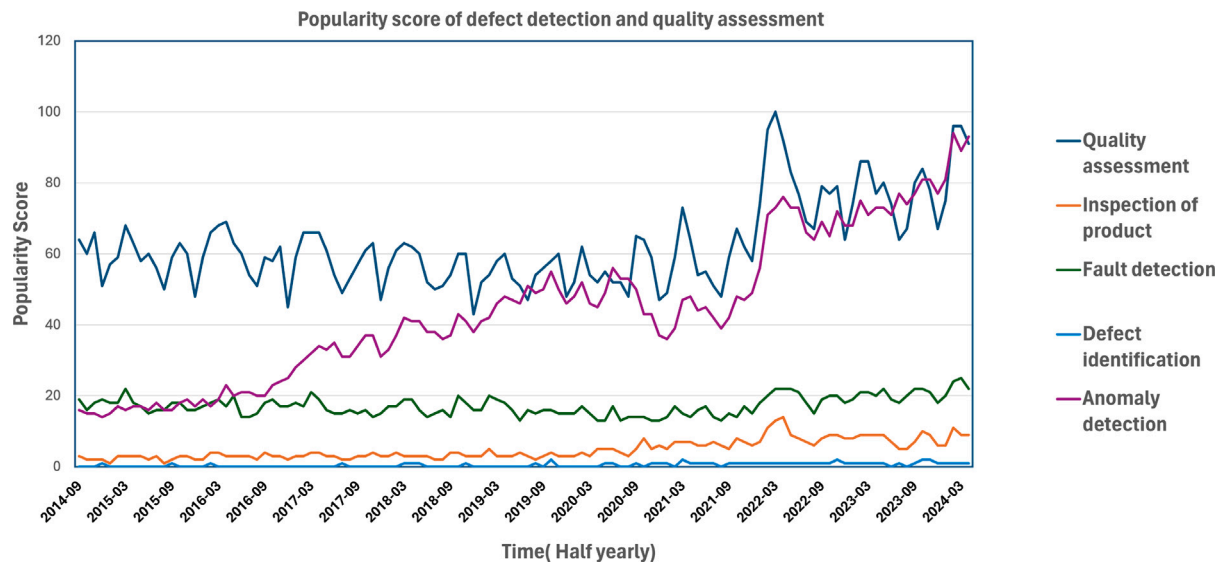


Fig. 2. The worldwide popularity score of product quality assessment in a range of 0 (min) to 100 (max) over time where x-axis represents the timestamp information and y-axis represents the corresponding score.

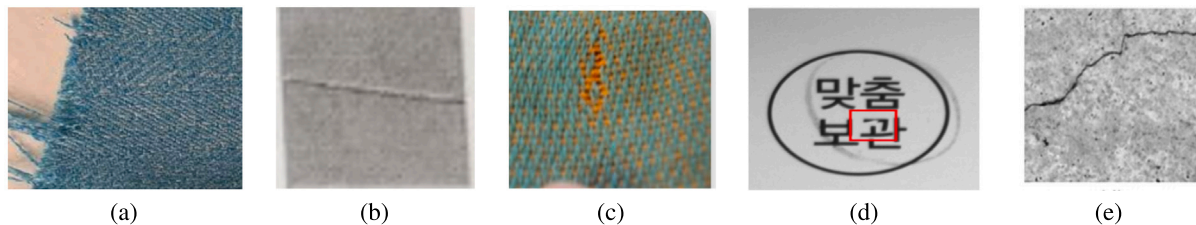


Fig. 3. Defect examples: (a) fray on fabric (b) crease on steel (c) colour imperfection on fabric, (d) silk screen printing and (e) crack on metal surface.

in the industry. Fig. 2 illustrates how research on quality inspection on industrial products is growing based on data collected from Google Trends [34]. So, by analysing the trend, it is seen that defect detection in industry is a growing field and quality assessment plays an important role to inspect and maintain the quality level of products in the industry.

Semi-automated methods work well for regular patterns but struggle with micro defects (e.g., cracks on metal). In contrast, machine vision offers more accurate detection and classification. However, the variability in defect shapes and sizes makes it difficult to classify all defects and build a comprehensive database. Fig. 3 shows examples of defects across different products. Henceforth, this section represents different categories of defects based on materials. The following subsections describe defect types, their characteristics, inspection methods and imaging technologies.

2.1. Defect categories and materials

This study focuses on several industrial products, made of metal, ceramic, textile, paper, wood and carbide, which may have various types of defects. Some defects are specific to specific products, for example, cracks can happen on products made of ceramic [35,36,48], metal or glass [8,45], but they do not happen on products such as art paper, fabric and trading card [38–40,49,50], which may have other specific defects like fray, colouring and printing. Printed products such as trading cards and art paper can have two types of defects. First, an art paper may have defects such as dents and fray on its edges and these defects cause damage directly to the product material (e.g., paper). Second, defects like pattern error and colour imperfection [47] are related to the printing quality of the art paper and they can happen during the painting or printing as each printing device has its own

unique printing limitations. In contrast, products such as tiles and glass have only the first type of defects (e.g., cracks and pinholes in tiles). Therefore, all possible defects in products are categorised into two: printing and non-printing. Table 1 represents various defects in these two categories along with the materials where the defects can happen. Printing defects like pattern errors, colour imperfection and poor line quality are visible on printed products (e.g., Pokemon cards, clothes, art papers). Non-printing defects are found in other industrial products, for example, steel, glass, and metal.

2.2. Defect characteristics

Defects can be of different shapes and provide different levels of human perceptions. Table 2 shows that defects are usually visible on three product locations: surface, edges, and corners. While defects on a product surface may have regular shapes (e.g., line, pattern), defects on product edges [8] and corners [35,36] often have random shapes. All defects (e.g., cracks, creases, colour imperfections) are visible to human eyes as this study focuses on the visibility and size of defects while estimating product quality. Some of the defects (e.g., cracks, creases) are palpable as they are perceptible with a finger touch.

Defects can happen for various reasons on aforementioned product locations. For example, cracks [7] may happen on edges and corners due to the movement of the product, while they occur on the surface during production due to high pressure on the product. Some defects (e.g., corrosion, contamination) may happen on the surface due to unfavourable weather conditions or chemical leakage. Poor construction of a product during manufacturing or assembling may be liable for these defects on the product surface when the product is in use.

Table 1
Defect categories based on materials.

Non-printing defects		Printing defects	
Defects	Materials or Products	Defects	Materials or Products
Chipping, crack	Glass [8]	N/A	N/A
Crack, pinhole, spot, blob, corner sharpness, glaze, scratch	Ceramic [35,36]	Printing error, torn, foreign substance, poor line quality	Art paper [37]
Fray, blob, hole, crease	Fabric [38–40]	Colour imperfection, pattern error, double printing	Fabric [40]
Chipping, dent, fray, scratch, corner sharpness	Trading card [41]	Spot, torn, discolour, foreign substance, centring	Trading card [41], art paper [37]
Shape error, chipping	Wood [42]	N/A	N/A
Sharpness, crack, crease, pit	Steel [43,44]	N/A	N/A
Crease, crack	Carbide [7]	N/A	N/A

Table 2
Characteristics and reasons for defects.

Locations	Shapes	Defects	Palpable	Visible	Materials /Products	Reasons
Edge	Random	Chipping	✓	✓	Glass, card [8,41]	Pressure
		Dent	✓	✓	Card, ceramic, steel [41,43,44]	Movement, pressure
		Fray	✓	✓	Card [41]	Faulty machine, usage
		Crack	✓	✓	Ceramic, steel, glass [8,43,45,46]	Friction, force
Surface	Line	Crack	✓	✓	Ceramic, glass, carbide [7,8,45,46]	High pressure, bad materials
		Crease, scratch, glaze	✓	✓	Steel, carbide, ceramic, glass [7,46]	Friction
	Pattern	Shape error	✓	✓	Card, wood [42]	Production error
		Discolouration	✓	✓	Card, textile, art paper [41]	Production error
	Spot	Pit	✓	✓	Ceramic, steel [43,44]	Mixing unwanted materials, pressure
		Spot, blob	✓	✓	Fabric [38,39]	Pressure, usage
	Patch	Corrosion	✓	✓	Steel, carbide [7,43]	Chemical leakage
		Contamination	✓	✓	Steel, ceramic [43]	Production error, poor construction
	Printing	Torn, foreign substance	✓	✓	Textile, art paper, trading card [37,47]	Old machine, usage, movement
Corner	Random	Sharpness, crack	✓	✓	Ceramic, glass [35,36]	Movement, pressure

2.3. Defects inspection

Defects inspection plays an important role as it involves checking the various attributes of a product and testing it to ensure that it meets specified standards. Defect inspection can take place in two levels, during manufacturing [8,43,44] and when the product is in-use [3,4,43,44,51–54]. Defects, which occur during the manufacturing of a product, can be inspected when the quality assessment of a product can be done by the company before bringing the product to the market. So, defect detection during manufacturing level is conducted on new products. When a product is in-use defect inspection can be conducted to estimate the status of the product. For example, one can buy a new Pokémon card from a company and use this card for some time before he/she sells this card. So, it is necessary to assess the quality of the card to estimate its current value. Table 3 represents different types of products and defects with their inspection levels.

Another attribute that has been estimated in both manufacturing and in-use levels is defect estimation (e.g., size, quantity). To check the product quality more deeply it needs to measure the quantity and length of the defect size. For example, the quality of the products made of wood has been assessed through some measurements such as length, quantity, and depth of defects on products while for other products, such as fabric, quality assessment has been done by length and quantity of the defects on products.

2.4. Defects imaging

Capturing the important product features and eliminating unexpected features (e.g., due to shadowing or improper lighting environment) are essential to maintaining the image quality that plays an important role in defect detection. To obtain the high-quality image of the product an appropriate image acquisition hardware (e.g., camera, scanner) with a good optical illumination platform is required. A suitable optical illumination platform overcomes the interference of environmental lighting, ensures the stability of the imaging system, and obtains an image with a high contrast.

Image acquisition hardware is chosen based on the product characteristics, defect types and sizes (e.g., small, very small) as well as the device characteristics (e.g., magnification, resolution). Table 4 represents descriptions of different devices along with their characteristics. High resolution and magnification allow for capturing more detailed information from the product that leads to an investigation of tiny defects which may not be visible to the naked eye. Thus, better performance in the process of detecting and classifying defects in general can be obtained using high-resolution images but they consume high computational resources.

Among the different types of image capturing devices [55–59] that have been presented in the literature, for example, on products made of semiconductor, if the defect size is very small (e.g., 1 to 10 nm) then a scanning electron microscope (SEM) [55] is used. If the defect

Table 3
Defect estimation and inspection levels.

Materials or Products	Inspection levels		Defects location	Defects estimation	Reference
	New	In-use			
Glass	✓	✗	Edge, surface	Length, quantity	Bukieda et al. [8], Hsu and Lin [45]
Tiles	✓	✓	Corner, edge, surface	Length, depth	Singh and Yadav [35,36], Matić et al. [48]
Fabric	✓	✗	Surface	Length, quantity	Chakraborty et al. [38], Ouyang et al. [39], Liu et al. [40], Almeida et al. [49], Liu et al. [50]
Trading card, art paper	✓	✓	Corner, edge, surface	Depth, quantity	Masod et al. [37]
Wood	✓	✗	Surface, edge	Depth, length, quantity	Chen et al. [42]
Steel	✓	✓	Surface, edge	Length, quantity	Chen and Li [3], Song and Yan [4], Psuj [43], Soukup and Huber-Mörk [44], Saiz et al. [51], Damacharla et al. [52], Altabay et al. [53]
Carbide	✓	✗	Edge	Length, quantity	Parenti et al. [7]

Table 4
Characteristics of different image capturing devices.

Devices	Defect size				Measurement		Products	Resolution (dpi)	Magnification	Reference
	VS	S	M	L	Ln	Dep				
Scanning Electron Microscope	✓	✓	✓	✗	✓	✗	Semiconductor	96–300	10 to 3,000, 000 times	Tomlinson et al. [55], Avinun-Kalish et al. [56], Becker et al. [57], Newell et al. [58], Jain et al. [59]
Digital Microscope	✓	✓	✗	✗	✓	✗	Card, glass, leaves	96–300	34 to 1,100 times	Bukieda et al. [8], Rolland et al. [60]
Digital Camera	✗	✓	✓	✓	✓	✗	All	300	3 to 5 times	Wang et al. [61], Nigam and Singh [62]
Photo-metric stereo	✗	✓	✓	✓	✓	✓	Steel, card, carbide	300	–	Saiz et al. [51]

Note: VS: Very small, S: Small, M: Medium, L: Large, Ln: Length, Dep: Depth.

size is small (e.g., 1 mm) or medium (e.g., 3 to 5 mm) then a high-resolution digital camera [61,62] is used for all products. In addition, for some products (e.g., woods, tiles) having deep defects (e.g., cracks) which need to be measured, photo-metric stereo [51] is used to capture images for depth measurement.

A brief description of each image capturing device is given below.

- **Scanning Electron Microscope:** An SEM [55–59] is used for defect detection on products made of semiconductors. It offers very high magnification, up to 3 million times, which allows very small, usually invisible defects, to be detected and analysed. Therefore, it provides high accuracy (e.g., 90% in [55]) when the captured data are processed with modern AI-based algorithms.
- **Digital Microscopes:** A digital light microscope [60] offering a magnification range of 34 to 1,100 times can capture images of various products while effectively detecting defects of different sizes, including those that are very small or small. For example, Rolland et al. [60] used a digital microscope to capture images of cotton leaves in their work to score the leaf hairiness which achieved 86%–96% accuracy. Bukieda et al. [8] also used digital light microscopy images to inspect the quality of glasses. They also described the importance of microscopic images to detect defects in edges and surfaces.
- **Digital Cameras:** A digital camera is used for all materials with small, medium and large defects. The usual photosensitive devices used in industrial cameras are charged coupled devices (CCD) or complementary metal oxide semiconductor (CMOS) chips. Hsu and Lin [45] uses two CCD cameras to capture images for glass quality inspection, while Nigam and Singh [62] used one CMOS camera to capture images for crack detection in beams.
- **Photo-metric Stereo:** To obtain images using photo-metric stereo [51], an object is sequentially illuminated by different light sources from different directions. It provides all information such

as contour lines as well as curvature on the surface, colour, spectral data and signal variation in horizontal and vertical directions. This approach is proficient in detecting defects of varying sizes, encompassing small, medium, and large, and possesses the unique capability to effectively identify deep defects as well. It can detect parts with diameters ranging from 65 mm to 110 mm and can detect breakage, bumps, and cracks. Saiz et al. [51] proposed a segmentation network to detect defects on metallic components using photo-metric stereo images which achieves 83% accuracy.

3. Classification of defect detection techniques

The literature covering defect inspection methods can be broadly divided into three parts: (i) manual, (ii) classical image processing and ML-based semi-automated and (iii) DL-based automated. For each defect inspection method, the advantages and disadvantages of visual inspection methods are presented in Table 5. Furthermore, semi-automated and automated inspection methods are discussed in the following subsections.

3.1. Semi-automated defect detection techniques

Semi-automated defect detection involves four key steps shown in Fig. 4: data acquisition, preprocessing, feature extraction, and classification. Data is captured via imaging setups described in Section 2.4 and enhanced through noise filtering. Features are then extracted using statistical, structural, filter, or model-based methods [18,63–65], followed by learning-based classification (e.g., SVM, KNN) [66,67].

The above mentioned approaches have been extensively researched in the literature. The choice of a method for defect detection depends on expert opinions, success rates, and dataset availability [49,50,68–70]. For instance, learning-based methods are effective for detecting defects in patterned surfaces, while statistical or model-based techniques

Table 5
Defect inspection methods.

Methods	Advantages	Disadvantages
Manual	Can identify diverse set of defects.	Time consuming, subjective, need experimental setup, costly, inconsistency, need human effort.
Classical image processing and machine learning	Lessen human errors and time.	Require skilled work for feature selection and extraction, limited algorithmic parameters, less accuracy.
Deep learning	Extract feature by themselves, lessen human efforts and high detection accuracy.	Require large dataset, poor calibration.



Fig. 4. Basic steps of an semi-automated defect detection system.

are suitable for identifying deformations on steel surfaces. Therefore, DL offers a solution by automating feature extraction, mitigating these limitations.

3.2. DL-based automated defect detection techniques

As shown in Fig. 1, DL-based defect detection is divided into two: classification and localisation. Classification models are trained either from scratch or via transfer learning using pre-trained weights (e.g., ImageNet) to reduce computational cost [71]. Localisation approaches vary by defect type and include encoder–decoder, pyramid, attention, and YOLO-based networks.

Henceforth, based on dataset availability and learning strategy, DL methods are classified into two paradigms: supervised learning and unsupervised learning. Both learning methods have been used for defect classification and localisation in many industries. In addition, semi-supervised, self-supervised, and weakly supervised learning have gained attention for reducing annotation effort while maintaining competitive performance. The basic definition about the model how they work are given below:

- **Supervised Learning:** Supervised learning has been the most widely used model with CNN for defect classification and localisation. Given a large training dataset, the supervised based models can attain substantial defect detection accuracy. However, collecting such a large labelled training dataset requires enormous manpower and makes the model expensive. This challenge can be addressed by unsupervised learning model.
- **Unsupervised Learning:** Unsupervised learning addresses the challenge of defect detection without labelled data, lowering annotation costs. Common approaches include autoencoders (AEs) and generative adversarial networks (GANs). AEs learn compact latent representations by reconstructing input images, while GANs generate realistic samples through adversarial training between generator and discriminator networks. Recently, diffusion models also have gained attention for their ability to iteratively denoise random noise into high-fidelity defect-free or synthetic images, offering superior reconstruction and anomaly localisation in complex textures.
- **Semi, Self, and Weakly Supervised Learning:** Semi-supervised learning (SSL) bridges the gap between supervised and unsupervised methods by leveraging a small amount of labelled data with a larger pool of unlabelled samples, offering good generalisation with reduced annotation cost. Self-supervised learning further minimises reliance on labels by learning from pretext task, such as contrastive learning showing strong results on small or imbalanced datasets. Weakly supervised learning uses coarse labels (e.g., image-level or sparse annotations) to reduce labelling effort while maintaining segmentation accuracy, though it may suffer



Fig. 5. Basic structure of a CNN network.

from lower precision and generalisability without attention or augmentation strategies.

- **CNN-based Networks:** As shown in Fig. 5, CNNs consist of multiple layers designed to extract high-level features from input images. Convolutional layers use filters to generate feature maps, and the network is trained using backpropagation to minimise a loss function. The final output layer performs classification at the image or pixel level [72,73].
- **Encoder–decoder Networks:** The encoder–decoder model is an established deep learning architecture utilised for defect segmentation. The input data is mapped to a lower-dimensional latent representation using the encoder, while the decoder takes this encoded form and generates the desired output. For example, U-Net model is a notable example of encoder–decoder models [74].
- **Pyramid Networks:** In defect detection for multi-scale analysis pyramid networks have been utilised. The concept of this approach involves extracting features from different resolutions, enabling a more comprehensive understanding of the image across various scales. The model can effectively learn by incorporating detailed information from both low and high-resolution features [75].
- **Attention-based Networks:** Recently attention mechanisms have gained significant attention in the field of computer vision. It enhances deep learning models by selectively focusing on important input elements, improving prediction accuracy and computational efficiency [76–79].
- **YOLO-based Networks:** Defect detection networks are divided into two types: one-stage and two-stage algorithms. One-stage models are YOLO [80] and SSD [81], predict object locations and categories in a single step and offers faster detection. Two-stage models, such as Faster R-CNN [82], first generate object proposals and then refine detection. Recently YOLO model and its several series gains more popularity in the defect detection.
- **Other Networks:** Beyond these networks, recent advanced networks include transformer-based models, foundation models, and large language models (LLMs) also gain more popularity. Transformers excel in capturing long-range dependencies and are effective for both classification and segmentation tasks. Foundation models, such as SAM and CLIP, offer strong generalisation with minimal supervision, while few-shot learning enables effective training on limited data. LLMs, when combined with vision models, enhance explainability and support multi-modal defect analysis through language-driven insights.

3.3. Architecture of automated defect detection methods

Fig. 6 illustrates a typical architecture of automated system for defect detection. Products move on a conveyor belt under suitable

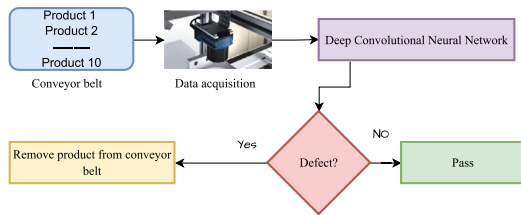


Fig. 6. Architecture of an automated defect detection system.

lighting, with industrial camera sets capturing image data from various angles and storing it on the industrial computer. Data acquisition depends on specific application requirements such as resolution, speed, and accuracy, as well as factors like lighting conditions and object size. The description of defect image-capturing devices for different types of products is provided in Section 2.4.

Common types of cameras for data acquisition include area scan cameras for 2D area imaging and 3D cameras for depth information for real-time applications. Moreover, scanners can also be utilised in certain applications where consistent lighting is crucial for defect detection. For instance, trading cards [83] can be scanned to detect defects while consistent lighting provided by scanners ensures uniform image quality, aiding in accurate defect detection. The visual system processing occurs on the computer, where modern machine vision models analyse the captured image to determine product defects. If a defect is detected, the system removes the product from the conveyor belt; otherwise, the product proceeds forward.

4. Efficacy of defect detection techniques using semi-automated learning

Machine vision is a popular approach for quality inspection of industrial products which utilises advanced computer vision algorithms and imaging technology to perform efficient inspections with reduced or without any manual assistance. Many studies have explored this approach, encompassing image processing and ML methods across various product domains. This section presents a comprehensive overview of these studies in two key aspects: image processing and machine learning.

4.1. Image processing-based semi-automated techniques

Image processing techniques [21,84,85] detect defects by identifying deviations from normal surface patterns, particularly when defects are visually distinct from the background. Common methods include edge-based, wavelet-based, and feature-based approaches [86, 87]. Edge detection techniques, such as Canny and Otsu are widely used; for instance, Parenti et al. [7] applies Canny edge for carbide inspection, detecting defects as small as 10–50 μm , while Bukieda et al. [8] uses Otsu thresholding to identify cracks and dark spots on glass with 80% and 60% accuracy, respectively. Wavelet-based methods enable multi-scale analysis. Lu and Chen [88] achieves 99% accuracy in detecting cracks on metal surfaces using wavelet-based segmentation, and Nigam and Singh [62] combines wavelet analysis with slope detection for beam crack identification. Feature-based techniques such as local binary pattern (LBP) combines with extreme learning machines (ELM) have shown 95% accuracy in cold-rolled steel strip defect detection [89], while colour and texture-based grading methods applies to ceramic tiles have achieved up to 96% accuracy [90,91]. Table 6 summarises the performance of these techniques, including device types and accuracy metrics. For instance, Singh and Yadav [35] reports 96% accuracy using image processing and mathematical operations on tile images, and Matic et al. [48] reaches 90% accuracy for edge and corner defects.

However, these above traditional image processing techniques are effective for detecting defects on uniform surfaces, they struggle in real-world scenarios with varying defect shapes, sizes, and textures. Extracting a fixed set of features limits adaptability across diverse products. To overcome these challenges, ML algorithms have been employed for their ability to manage such variability within a scalable and robust framework.

4.2. Classical machine learning based techniques

In this section, various classical ML algorithms that include classification, regression and clustering are discussed [94,95]. They can be categorised into three major groups based on their learning strategy: supervised learning, unsupervised learning, and semi-supervised learning. Another technique called the hybrid method is also used to increase the performance in learning. Table 7 summarises the accuracies for these ML algorithms across studies.

4.2.1. Supervised learning

Supervised learning [69,111] is commonly used for defect classification using labelled data, where models are trained on known datasets and validated on unseen data. Support Vector Machine (SVM) has shown strong performance in applications, such as laser welding, copper strip inspection [66,101,102], and steel defect classification in aerospace and automotive domains. [111]. Mandriota et al. [99] uses SVM with gabor and wavelet features for rail defect detection, where gabor filters outperformed others. The K-Nearest Neighbour (KNN) algorithm achieves 96% accuracy in fabric defect detection using grey-level co-occurrence matrix features [96], and is also applied to gear crack detection with feature selection [97]. Random forest and logistic regression, combined with interpretability tools, is used by Oliveira et al. [104] for tire defect detection with 96% accuracy. Zorić et al. [112] applies fourier spectrum annuli for tile defects, achieving a 99% F1-score. A fused discrete cosine transform and discrete wavelet transform-based ANN is employed by Hemalatha et al. [105] for textile inspection, while Anupriya et al. [106] uses one-class SVM to achieve over 95% accuracy. In steel surface inspection, Dorbane et al. [108] applies synthetic minority over-sampling technique (SMOTE) with ensemble methods, XGBoost, for robust classification. Principal component analysis (PCA) with SVM is used by Li et al. [107] to classify reflective steel ball defects with 97.15% accuracy. Finally, the Naive Bayes (NB) classifier achieves over 98% accuracy in steel surface defect classification [5], highlighting its lightweight and efficient performance.

4.2.2. Unsupervised learning

Unsupervised learning algorithms for defect detection work on unlabelled data and often rely on clustering methods to identify patterns. K-Means clustering, which groups data into K clusters based on similarity, has been effectively applied to defect detection tasks in semiconductor materials, fabrics, and machine fault diagnosis [98,100,113], achieving up to 95% accuracy in fabric defect detection [100]. Another widely used approach is the Self Organising Map (SOM), which maps high-dimensional inputs to a lower-dimensional space while preserving topological relationships. SOM has been employed for detecting wood surface defects using colour and texture features [114–116], with Silvén et al. [103] applying it to identify defects on lumber boards. Although SOM-based methods yield promising results, they typically require a sufficient number of defect samples to achieve high accuracy.

4.2.3. Hybrid methods

To improve the performance of defect detection and classification, hybrid methods integrate two or more ML techniques. This have been applied in many research: the combination of the SOM and the SVM is used to detect defect patterns on wafers [117], combination of the genetic algorithm and the SVM is used for detecting defects in

Table 6
Summary on defect detection based on image processing.

Defects and products	Dataset size	Image processing type	Method	Accuracy
Edge defects on bridge [92]	Small	Edge-based	Line inspection	Detect 1-4 nm wide cracks.
Surface defects on metal [88]	6204	Wavelet-based	N/A	99.5%
Surface defects on beam [62]	Small	Edge-based	Canny	Detect crack of 4 mm depth.
Surface defects on steel strips [89]	N/A	Feature-based	LBP	95%
Surface defects on concrete [93]	3420	Edge-based	Canny	53%–79%
Edge defects on glass [45]	Small	Th-based	Otsu	Fracture: 60%, crack: 80%
Corner defects on tiles [35]	100	Edge-based	Canny	96.524%
Corner defects on tiles [36]	N/A	Edge-based	Pythagorean theorem	90%
Surface defects on Carbide [7]	Small	Edge-based	Canny, contour	N/A
Corner defects on tiles [48]	100	Th-based	Contour	90%

Note: Th: Threshold, N/A means not appropriately defined.

Table 7
Summary of ML techniques in defect detection.

Objects	Feature extractor	Classifiers	Dataset size	Accuracy
Fabric [96]	GLCM	KNN	400	96%
Gear [97]	Two stage feature selection	Weighted KNN	N/A	N/A
Engine [98]	Genetic algorithm	k-means	Small	94%
Rail [99]	Gabor filter, wavelet method	SVM	400	N/A
Fabric [100]	K-Means	K-Means	160	95%
Coopers strips [101]	Wavelet Transform	SVM	N/A	96.7%
Wood [67]	GLCM	SVM	355	94.37%
Wood [67]	LBP	SVM	355	95.78%
Steel [102]	PCA	SVM	13 925	81%
Steel [5]	DCT, SVD	Naïve Bayes'	500	98%
Wood [103]	LBP	SOM, KNN	42	80%
Tire [104]	RF, LR	RF, LR	107	RF: 92%, LR: 96%
Fabric [105]	GLCM-DCT, ANN	GLCM-DCT, ANN	1067	80%
Fabric [106]	One class SVM	One class SVM	200–2000	>95%
Steel [107]	PCA, K-CV, SVM	SVM	214	97.15%
Steel [108]	XGBoost, SMOTE	XGBoost	1941	99.16%
Steel [109]	DenseNet	ML	1800	99.72%
Steel [110]	AlexNet, KNN, LDA	AlexNet, KNN, LDA	1800	Up to 100%

Note: SVM: Support Vector Machine, KNN: K-Nearest Neighbour, SOM: Self Organising Map, PCA: Principal Component Analysis, GLCM: Grey Level Co-Occurrence Matrix, LBP: Local Binary Pattern, DCT: Discrete Cosine Transform, SVD: Singular Value Decomposition, RF: Random Forest, LR: Logistic Regression, SMOTE: Synthetic Minority Oversampling Technique, LDA: Linear Discriminant Analysis.

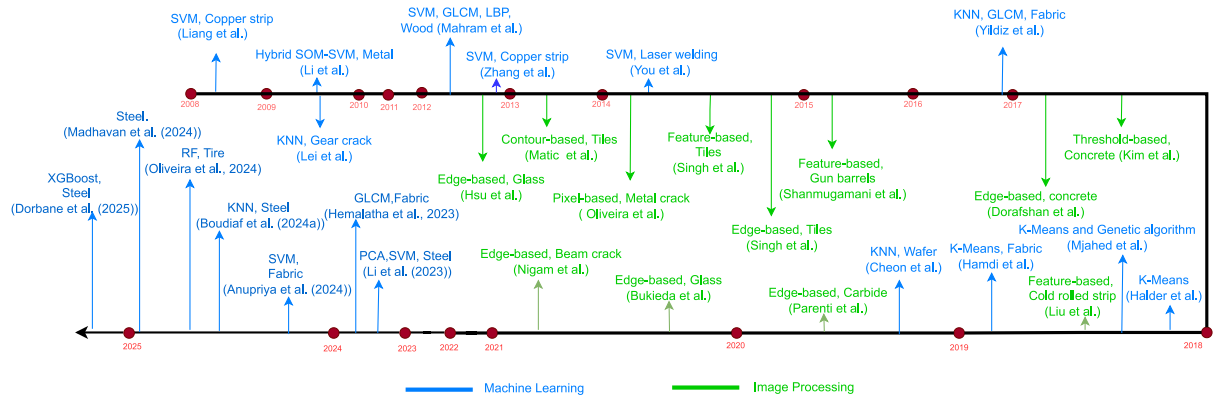


Fig. 7. Chronological overview of defect detection on different products using image processing and classical machine learning.

aluminium [68] while the combination of KNN and CNN is used to identify unknown class defects on wafers [69]. Madhavan et al. [109] integrates features extracted from DenseNet201 with tree-based classifiers, reaching 99.72% on the NEU surface defect dataset. A chronological overview of the most relevant method of defect detection on different products using image processing and ML is shown in Fig. 7. One of the advantages of ML algorithms is their ability to achieve good results even with limited sample data. However, a limitation of these algorithms is their reliance on explicitly engineered features, which can be challenging to acquire in complex cases. In real-world conditions, these methods perform excellent under some specific conditions (e.g., parameter setting) but are sensitive to variations. In

contrast, DL-based model overcomes these limitations by automatically extracting features and classifying defects, requiring only ground truth data. It also offers greater adaptability across various defect types and applications.

5. Efficacy of DL-based automated defect detection

DL is rapidly advancing in the AI field for defect classification and localisation. Numerous studies [20,43,44,118–120] have applied DL across various product domains. This section critically reviews the use of DL in defect inspection covering both supervised and unsupervised methods for classification. It also examines segmentation and detection

Table 8

Advantages and disadvantages of defect classification and localisation networks.

Learning strategy	Defect inspections	Models	Advantages	Disadvantages
Supervised Learning	Classification [49,50,70,119,121–123]	Pre-trained model (VGGNet, Alexnet etc.), CNN.	Require only image-based labelling; has a high learning capability; good classification accuracy; high speed training.	Cannot detect defect location.
	Localisation [46,47,74,124–127]	Semantic Network (FCN and U-Net).	Can detect tiny defective areas.	Requires of pixel-level annotation and time-consuming.
Unsupervised Learning	Classification [128–130]	CNN with GAN.	Can learn from unlabelled or partially labelled data; no need for annotation.	Need to comprise in detection accuracy; cannot find the defective area.
	Localisation [24,131,132]	Hybridised model, CycleGAN.	No annotation is needed at the pixel-level; can be used in real-time application.	Training and validation are slow.

Note: FCN: Fully Convolutional Network, GAN: Generative Adversarial Network.

techniques for localisation, including encoder–decoder, pyramid, and attention-based networks for segmentation, and pyramid and attention-based YOLO models for detection. Table 8 summarises the strengths and limitations of these approaches. Beyond these, semi-supervised, weakly supervised, and self-supervised learning approaches are also gaining attraction for reducing annotation demands while maintaining high performance. Additionally, advanced architectures such as transformers, foundation models, few-shot learning, and LLMs are increasingly being integrated into defect inspection pipelines, enabling generalisation and interactive defect analysis.

5.1. Defect classification methods

The goal of defect classification network is to classify different types of defects on the surface of the product. The most popular model is supervised learning, which can achieve significant defect detection accuracy when given a large training dataset.

5.1.1. Supervised learning (CNN-based methods)

Supervised CNN are considered as popular methods for both defect classification and localisation. They have been applied in various products such as on LED chips (95% accuracy [119]), laser welding in battery (99.87% accuracy [122]), fabric (95% accuracy [49], 98.1% accuracy [50]) and fabric print [38]. Park et al. [121] employs a supervised DCNN model to detect scratches on mobile phone screen covers and achieve an accuracy of 99% using a dataset of 16,800 labelled images. Other supervised DCNN model is employed by Dorafshan et al. [93] for crack detection on concrete and achieved 99% accuracy. Another work for crack detection on concrete has been done by Cha et al. [133] without calculating the defect features. Their method achieves good accuracy. Moreover, crack detection on concrete using DCNN model by Golding et al. [134] achieves 99% accuracy.

Ouyang et al. [39] proposes a system where they applies image pre-processing, fabric motif determination, candidate defect map generation and CNN to determine the defects in the fabric. Their method detect defects with 98% accuracy. Almeida et al. [49] introduces a cost-effective method for defect detection on fabric with high accuracy. They develop a new CNN model from scratch and incorporate a false negative reduction technique to enhance the performance of the system. Their system achieves an accuracy of 95%. Chakraborty et al. [38] proposes a DCNN method for identifying the spot (e.g., printing defect) and mismatch print on fabric. Tabernik et al. [135] proposes a two-stage surface inspection model combining a segmentation network with a DCNN for defect classification. Similarly, Ren et al. [136] uses a pre-trained CNN to extract patch-based features for defect classification. In order to deal with erroneous annotation data and image uncertainty, Yu

et al. [137] presents a DCNN model to handle label noise added by human annotators. In their work, noise coming from the surroundings is eliminated by using a symmetric residual filtering module. Then a combined probabilistic classification model is trained using the improved dataset to identify surface flaws in PCBs. He et al. [138] also proposes a DCNN method combined with a multilevel feature fusion network on steel surface to classify the defects. Their experimental result shows that it improves accuracy than other CNN.

5.1.2. Supervised learning (Transfer-learning-based models)

In the pretrained model, it is difficult to configure a model by choosing the depth of the model and Kernel size. To configure a defect detection model, Konovalenko et al. [139] explores 14 neural network models for defect detection on the surface of the metal. They implement various models including InceptionV3, Inception-ResNetV2, ResNet, DenseNet, EfficientNet, MobileNet and U-Net to detect defects. Their study reveals that DenseNet and ResNet models perform very well with a depth of 50 layers, and exhibit the strongest generalisation capabilities for defect detection.

Singh and Desai [140] reduces computation for steel defect detection by extracting features with ResNet-101 and classifying them using SVM, improving results through data augmentation despite limited samples. For fabric inspection, Jun et al. [23] proposes a two-step learning-based method involving local and global defect prediction. They augment data, enhance images via histogram equalisation, use Inception-V1 for local prediction, and LeNet-5 for final classification. Liu et al. [50] modifies VGG16 to improve recognition of complex textile defects, achieving 98.1% accuracy and faster processing.

In the industry perspective, sometime sophisticated feature extraction and modelling techniques are needed to distinguish between small variations and real faults. Mewada et al. [141] proposes a CNN approach based on residual networks to improve fabric defect classification. ResNet50, a pretrained residual network, is refined to categorise fabric defects into four groups. The method achieves 95.36% accuracy.

These both methods provide significant defect classification accuracy in diverse patterns of defects. However, it is a great challenge to collect a comprehensive representation of all possible defect patterns. This leads to imbalanced datasets where certain categories may have limited samples, making it challenging to adequately labelled all data. To address this limitation unsupervised model prove to be advantageous.

5.1.3. Hybridised models

Hybrid models exhibit increased performance by combining different approaches in defect classification. For example, Li and Zhang [144] proposes a hybrid model where the Gabor filter is combined with

Table 9

Summary of some contributions on defect classification using supervised learning. Note: TL: Transfer Learning.

Reference	Object	Image/Database	Method	Accuracy	Func.
Park et al. [121]	Glass	16,800	CNN	Acc: 99%	CNN
Dorafshan et al. [93]	concrete	–	CNN	Acc: 99%	CNN
Cha et al. [133]	Conc- rete	332	CNN	Acc: 98.22% (51 epochs) and Acc: 97.95% (49 epochs)	CNN
Golding et al. [134]	Conc- rete	40,000	CNN	F1-score: 99.331% (10 epochs) and 99.549% (20 epochs)	CNN
Ouyang et al. [39]	Fabric	1,160	CNN with pairwise potential activation	Public dataset: 98%, manual dataset: 90%	CNN
Almeida et al. [49]	Fabric	TILDA, MVTec	CNN with false negative reduction	Acc: 75%	CNN
Saiz et al. [51]	Steel	> 25,000	CNN	Low processing time	CNN
Chakraborty et al. [38]	Fabric	380	CNN	Acc: Defect free: 56%, Spot: 73%, Misprint: 62%	CNN
Tabernik et al. [135]	steel	KolektorSSD	CNN with segmentation	AP: 95%	CNN
Ren et al. [136]	wood	NEU-DET	CNN with pixel-wise segmentation	Acc: 94.29%	CNN
[137]	PCB	HRIPCB	Auxiliary inference model was used to manage the noise on labels	Acc: 79.62%,	CNN
He et al. [138]	steel	NEU-DET	CNN with multi feature fusion	Acc: 99.67%	CNN
Zheng et al. [142]	Steel	DAGM, NEU, CCL	CNN	Acc: DAGM: 99%, NEU: 99.95%, CCL: 95.67%	CNN
Altabey et al. [53]	Pipe- lines	900	CNN	Acc: 93.53%, Regression rate: 92.04%	CNN
Xu et al. [143]	Bridge	2,068	CNN, depth-wise separable model	Acc: 96.37%, Specificity: 95.83%, F1-score: 87%	CNN
Singh and Desai [140]	Steel	–	ResNet-101 +SVM	AP: 100%	TL
Jun et al. [23]	Fabric	5,000	InceptionV1, LeNet-5	Acc: 92%	TL
Liu et al. [50]	Fabric	6,000	DCNN, VGG16	Acc: 98%	TL
Mewada et al. [141]	Fabric	TILDA, DAGM	ResNet50	Acc: 95.36% , F1-score : 95.35%	TL
Li and Zhang [144]	Fabric	–	Gabor-filter, CNN	Acc: 98.6%	Hybrid
Chen et al. [42]	Wood	> 45,000	WDD-DL system, Inception ResnetV2	Precision: 97%, Recall: 90% and F1-score: 92%	Hybrid
Damacharla et al. [52]	Steel	12,568	Transfer learning (TLU-Net)	Perform 5% better than base model	Hybrid
Dharmawan et al. [145]	metal	–	Ensemble CNN	Acc: 99.03%	Hybrid
Zhou et al. [146]	Steel, Glass	NEU-DET, DAGM-2007	Double-VGGNet	Acc: NEU-DET: 99.44%, DAGM-2007: 99.49%	Hybrid

a neural network to detect non-printing defects on fabric and achieves 98.6% accuracy. Chen et al. [42] proposes a hybrid model to classify defects (e.g., high, normal, overlong, and short) on wood surfaces. The suggested WDD-DL system incorporates various techniques for pre-processing in the first stage and various DCNN models in the second stage to detect defects. Damacharla et al. [52] employs a CNN model for automatic steel defect detection and they further enhance the performance by utilising a hybrid method called TLU-Net. Dharmawan et al. [145] uses an ensemble DL technique for defect shape classification with good accuracy (e.g., 99.03%). In another work, a novel defect detection network employs for surface inspection where features are extracted using labelled double-VGG16 [146]. It provides the highest detection accuracy but is time-consuming.

Altogether hybrid model provides good accuracy but it is a slow training process and requires a substantial amount of data. Moreover, the proposed model has some limitations, especially on tasks such as the localisation of defects in complex textures. Table 9 offers a comprehensive overview of the key aspects related to defect classification in various articles using supervised technique. The table provides information on the achieved accuracy of various methods, the dataset size utilised in the experiments, and the methodologies employed by each of the respective studies.

5.1.4. Unsupervised learning

Most recognised models for unsupervised learning are autoencoder-based (AE) and GAN-based model. For example, Zenati et al. [147] trains a GAN-based model for defect detection more efficiently on high-dimensional datasets. Lian et al. [128] proposes a hybrid model combining CNN and GAN to detect small surface defects. GANs have also been used to generate synthetic data for surface inspection tasks [129, 130]. To enhance detection speed, the fast AnoGAN [148] is introduced, though such methods still struggle with complex patterns. A CycleGAN-based method [149] is used to annotate defective pixels and train a U-Net model, demonstrating promising results across various surface types. Yu [150] proposes an improved AE-based feature learning method for wafer defect detection, achieving 97% accuracy. Mujeeb et al. [151] applies AE for IC defect recognition with 85% sensitivity and 7% specificity, enabling training from a single sample via data augmentation. Haselmann et al. [152] reports 98% AUC for unsupervised surface inspection of plastic parts. Shakir and Topal [153] introduces an unsupervised spectral domain method for fabric fault detection using patch-based spectral comparison, reaching 94% accuracy and contributing a dataset of ten fabrics with 27 fault types. Tian and Li [154] uses AE-based pixel-wise classification by encoding/decoding

Table 10

Summary of some contributions on defect classification using unsupervised learning.

Reference	Object	Image/Database	Method	Accuracy	Function
Lian et al. [128]	Steel	–	CNN using local variation with GAN	Acc: 99.2%	GAN
Niu et al. [129]	Multi-object	CCSD-NL	SDGAN	Error rate reduce from 3.5% to 1.77%	GAN
Deecke et al. [130]	Multi-object	–	GAN	Acc: 94%	GAN
Tsai et al. [149]	Wood	400 images	CycleGAN with input–output pairs and U-Net	Acc: 99.49%	GAN
Yu [150]	Wafer	WM-811K	Enhanced stacked denoising autoencoder with manifold regularisation	Acc: 89.6%	AE
Mujeeb et al. [151]	IC	–	One class-based feature with AE	Sensitivity: 85%	AE
Haselmann et al. [152]	Plastic parts	–	One class-based feature with AE	AUROC: 98%	AE
Shakir and Topal [153]	Fabric	Ten fabric dataset	local spectra with AE	Acc: 94%	AE
Tian and Li [154]	Fabric	TILDA	Cross patch similarity with AE	Acc: 99%	AE

Note: GAN: Generative Adversarial Network, AE: Auto Encoder.

patches with latent variables, improving defect localisation by adjusting the latent space. Table 10 presents the summary of the aforementioned papers with their respective applications and functionalities.

5.1.5. Discussion

A common characteristic observed in this category is the predominant use of supervised learning approaches, which necessitate labelled data for the training stage which is the main challenge in this category. Unsupervised learning significantly reduce the effort required for labelling and manual annotation of image datasets at the pixel level, as they do not rely on a large number of labelled defective sample. The primary drawback of unsupervised models is their lower accuracy and reliability compared to supervised learning.

5.2. Defect segmentation techniques

Another challenge in defect localisation is defect segmentation, which is utilised to assess, locate and quantify flaws. These elements are also necessary for a thorough evaluation of product quality [124, 125, 155, 156]. Therefore, in this section, several research articles are highlighted that have employed DL-based segmentation techniques for defect localisation in these aspects. Table 11 presents the key features of papers in the field of encoder–decoder, pyramid network and attention-based categories for defect segmentation.

5.2.1. Encoder–decoder based supervised segmentation

Supervised segmentation models, a fully convolutional network (FCN), is initially proposed with effective learning [157]. By using this concept of FCN, Balzategui et al. [158] develop a FCN architecture to generate a defect segmentation map and identify the region of the defect accurately. Zambal et al. [126] uses semantic segmentation technique to implement an automated fibre placement defect segmentation. Liu et al. [125] also uses semantic segmentation for defect detection on fabric with 99.6% accuracy. Later, Ronneberger et al. [74] proposes another semantic segmentation model, U-Net, which is used by several researchers in their studies for diverse sets of applications and obtain efficient results. By modifying the U-Net model and combining with regional proposal networks (RPN), Han et al. [127] proposes a hybrid approach to identify defect regions on silicon wafers. In another research [124], a new end-to-end trainable model based on segmentation network is proposed for crack detection using multi-scale feature learning. Their proposed method outperforms the U-Net model with low cost and good accuracy. For surface defect detection, Lu et al. [159] suggests a real-time segmentation-based method where MobileNetV2 is used to extract features then reverse attention

modules are used to focus on the key region. Finally, a boundary refinement module is used to detect defects. Regarding generalisability the suggested technique performs well but needs annotated datasets.

Drass et al. [46] uses semantic segmentation with AI-generated masks to detect glass edge cracks, achieving 99% accuracy. Jing et al. [156] combines MobileNetV2 with semantic segmentation for fabric defects, while Ning et al. [160] applies a U-Net variant with multi-scale sensing for ceramic defects, reaching 95.13% mAP but depending heavily on mask accuracy. Bai et al. [161] enhances steel defect segmentation with group normalisation and split-attention, achieving 89.11% IoU. Neven and Goedemé [54] proposes a multi-branch U-Net for defect classification, though it struggles with foreground-background confusion. To capture broader context, Wang et al. [162] introduce a hybrid transformer model. Luo et al. [163] present I2GF-Net for complex textures with semantic enhancement but high computational demand. Chen et al. [164] develops A-Net for lightweight real-time segmentation, with limited success on diverse defect types. Zhu et al. [165] uses ResNet34 with depth-wise separable convolutions to reduce redundancy in wood defect detection. Moreover, there are some contributions on the pixel level segmentation have been done in the last few years. He and Liu [166] proposes a pixel-level segmentation model on the DAGM dataset and achieves an MIoU of 0.845. Another segmentation model named TAS2-Net is implemented by Liu and He [155] with the same dataset, increased MIoU of 0.869. To find the defective region, Cao et al. [167] also employs a pixel-wise segmentation network using multilevel feature module in the RSDD dataset and achieves 0.85 in terms of mIoU.

Encoder–decoder model, FCN and U-Net, deliver strong segmentation but faces practical limitations in industry due to reliance on dense annotations, sensitivity to class imbalance, and difficulty handling small or low-contrast defects. Lightweight variants improve speed but often reduce accuracy. Enhancements, for example, pyramid networks and attention modules improve multi-scale learning and defect focus, but it still add computational complexity. In most of the research works surveyed, it has been shown that U-Net performs better than FCN for segmentation networks. Pixel-level segmentation has also achieved significant success in defect segmentation.

5.2.2. Pyramid network-based supervised segmentation

These methods focus on improving results through the use of multi-scale features. They obtain rich contextual information from lower scales and high resolution from the largest scale. However, challenges related to real-time processing are the primary issues in this category.

Lin et al. [168] addresses the issue of identifying small, low-contrast flaws using ResNet-34 as backbone and combine with global dynamic

Table 11

Summary of some contributions on defect segmentation using DL. Note: GAN: Generative Adversarial Network, ED: Encoder–decoder, Attn: Attention, FCN: Fully Convolutional Network, GAN: Generative Adversarial Network.

Reference	Object	Images/Dataset	Model	Accuracy	Function
Liu et al. [125]	Fabric	DAGM 2007	Modified semantic segmentation	Acc: 99.6%	ED
Ronneberger et al. [74]	Silicon wafer	106 images	Region Proposal Network (RPN), UNet	Precision: 83%	ED
Zou et al. [124]	Multi-object	260 images	SegNet	F1-score: 87%	ED
Drass et al. [46]	Glass	—	Unet + Xception	IoU: 99.25%	ED
Jing et al. [156]	Fabric	YFI and FI	MobileNetv2, semantic segmentation	Precision: 99%, F1-score: 95%	FCN
Ning et al. [160]	Ceramic		Semantic segmentation (U-net)	mAP: 95.13%, IoU: 90.70%	ED
Bai et al. [161]	Steel	kaggle	Encoder–decoder + split-attention	IoU: 89.11%	ED
Neven and Goedemé [54]	Steel	10K images	Hybrid network	mIoU: increase of 6.8%	ED
Wang et al. [162]	Fabric	SD-saliency-900, fabric	Hybrid transformer with CNN	mAP: SD-saliency-900 dataset (97.91%), fabric (98.08%)	ED
Luo et al. [163]	Wood	VSb-DET, NEU-DET and OULU-DET	Information guidance feedback + top-down feedback encoder + semantic segmentation	mAP: VSb-DET: 78.4%, OULU-DET: 63.6%, NEU-DET: 83.9%	ED
Chen et al. [164]	Steel	NEU, DAGM and MT	A-shape lightweight neural network	IoU: DAGM: 82.86%	ED
Zhu et al. [165]	Wood	1,100	Depthwise separable dilated conv	IoU: 79.73%, F1-score: 88.11%	ED
He and Liu [166]	Steel	DAGM	Pixel-level segmentation	mIoU: 84.5%	ED
Liu and He [155]	Steel	DAGM , NEU-DET	TAS2-Net	mIoU: 86.9%	ED
Konovalenko et al. [139]	Metal	> 1.4 million	ResNet, DenseNet, U-Net	Acc: 93% and IoU: 91%	ED
Yoon et al. [47]	Art paper	349	Improved U-Net++	Dice score: 72%	ED
Cao et al. [167]	Steel	MT, RSDD, and CFD	Pixel-level	mIoU: (MT:73.7%, RSDD: 85%, CFD: 60.1%).	ED
Lin et al. [168]	Tiles, PCB	DAGM, MT	Multi-scale fusion feature, and local pyramid edge feature through Resnet34.	IoU: DAGM: 86.33%, PCB: 95.31%	Pyramid
Meng et al. [169]	Tiles	KolektorSDD	Semantic segmentation networks+ spatial pyramid pooling	IoU: 61.97%, Acc: 99.51%	Pyramid
Liu et al. [170]	Rail surface	–	Pyramid feature extraction+CNN	Ap: 81.41%	Pyramid
Wang et al. [79]	Chip	–	CSPDarkNet53-SA and spatial pyramid with path aggregation network	mAP: 78.47%	Attn
Liu et al. [76]	Tiles	NEU-DET, DAGM	Multi shuffle-block dilated convolution with dual attention	AP: 91.47%, IoU: 89.52%, Acc: 99.25%	Attn
Zhong et al. [171]	Wood	–	Deeplabv3+ positional Attn	IoU: 82.79%, F1-score: 83.90%	Attn
Luo et al. [78]	Steel	DAGM, KolektorSDD	Saliency heatmap memory attention	AP: 96.2%	Attn
Yang et al. [172]	Multi-object	SD-saliency-900, RSSD	Pixel level with attention	Dice Coefficient: 85.4%	Attn
Shao et al. [132]	Fabric	FID, AITEX, TILDA	Pixel-level segmentation with multitask mean teacher	mIoU: 82.5%	Attn
Niu et al. [173]	Multi-object	DAGM, KOLEKTORSDD	GAN + attention module	AP: 99.24%, IoU: 59.43%	GAN
Kim et al. [24]	Metal	–	PixelGAN	–	GAN
Xu et al. [174]	Fabric	11 800	GAN	mAP: 78%	GAN
Wan et al. [175]	Fabric	AITEX, TILDA, MVTEC	Unsupervised high-frequency feature mapping	AUC: improvement of 5.3% in detection and 2.9% in segmentation	–

convolution and multi-scale fusion features to extract spatial feature. Then, these features are combined with reverse attention modules to improve the model performance in identifying small surface defects. In order to achieve accurate results with a small number of training examples, Meng et al. [169] presents a semantic segmentation network with spatial attention for surface defect identification. The network uses a double branch architecture made up of a detail-semantic guide module, coupled spatial pyramid pooling, and a modified residual shrinkage module. The goal of the suggested framework is to minimise the need for a sizable annotated dataset for surface defect detection. Liu

et al. [170] also uses pyramid network to extract multi-scale feature maps then these feature are given as an input into a lightweight CNN model with a limited number of parameters for segmentation.

These architectures often involve complex designs and high computational costs, which hinder their suitability for real-time or resource-constrained environments. Despite efforts to reduce annotation requirements and parameter sizes, such as using lightweight backbones or spatial attention, the integration of multiple modules can still increase model latency and training complexity. Balancing accuracy with efficiency remains a key challenge for broader industrial adoption.

5.2.3. Attention-based supervised segmentation

Attention mechanism allows the model to focus on specific sections of the input while executing a task. It dynamically assigns weights to different elements in the input, indicating their relative importance or relevance.

For example, Wang et al. [79] enhances tiny defect detection in semiconductors using CSPdarkNet53-SA with spatial attention, spatial pyramid pooling, and a path aggregation network to capture multi-scale fine-grained features. Similarly, Liu et al. [76] proposes an adaptive segmentation network that combines multi-scale defect features with dilated convolution and dual attention context guidance, effectively identifying small and texture-similar defects in real industrial datasets.

Recently, researchers have been increasingly focusing on improving pixel-level segmentation by incorporating attention mechanisms. For example, Zhong et al. [171] designs a deep Gaussian attention network for segmenting timber defects using Deeplabv3+. This network improves segmentation accuracy even in difficult locations by utilising a transformer-based attention mechanism to capture global information. The network handles ambiguous defects and different forms and sizes of defects by utilising astrous spatial pyramid pooling and incorporating a Gaussian module into attention mechanisms. Luo et al. [78] also develops a model where a feature pyramid network is used to extract features. Convolutional operators and memory attention-enhanced feature modules are used to generate a saliency map. Then, the features from saliency map are used to conduct prediction. The framework works excellently using a small number of samples with pixel-level annotation. Another pixel-level segmentation network is deployed by Shao et al. [132] on fabric and achieved 0.825 of MIoU. Due to a minimal portion image pixels of defects, the pixel-level defect detection task is always against class-imbalance issue which also will affect the detection performance. To address this issue Yang et al. [172] proposes an attention-based segmentation network to detect welding defects with a dice coefficient of 0.854.

These aforementioned methods perform excellently in defect localisation but still face problems in detection accuracy and data imbalance. Attention-based segmentation enhances defect detection by focusing on relevant regions, improving localisation of small or ambiguous defects in complex textures. Spatial, channel, and transformer-based attention modules boost contextual understanding and have been effectively integrated into models like U-Net and Deeplabv3+. However, their benefits come at the cost of increased complexity, memory use, and sensitivity to domain shifts and class imbalance, limiting real-time deployment unless carefully optimised.

5.2.4. Defect segmentation by unsupervised learning (GAN-based)

This section highlights some studies on unsupervised or weekly supervised model. GAN-based models, such as cycleGAN and pixelGAN, are very widely used unsupervised model for defect segmentation. For example, Niu et al. [173] proposes a weakly supervised defect identification technique that locates and segments defect areas using image level annotations. To do this, they combine a defect attention module with CycleGAN. The generator is trained with test images for flaw detection through comparison using the suggested method. The experimental findings show excellent defect segmentation performance across a range of industrial applications. These approaches offer valuable alternatives for detecting surface defects without relying heavily on labelled data. Kim et al. [24] employs a pixelGAN-based model on semiconductor manufacturing data to identify the defect and segment the defective area. The comparison result shows that their method performs better than another base model in a real industry dataset.

Xu et al. [174] addresses limited training data in fabric defect detection by using GAN-based augmentation. By combining defect localisation maps, a dual U-Net, and multi-head attention, the model significantly improves accuracy across five defect types—float, line, knot, hole, and stain from 41%, 44%, 38%, 42%, and 41% to 78%, 76%, 72%, 67%, and 64% respectively. Wan et al. [175] proposes an

unsupervised high-frequency feature mapping model that captures rich high-frequency details and uses a query-key attention module for defect comparison. Their approach improves average AUC by 5.3% in detection and 2.9% in segmentation, outperforming prior state-of-the-art fabric methods.

5.2.5. Discussion

All studies in this category focuses on segmenting defects in images. The encoder-decoder based networks has the highest percentage of deep learning in the literature. This indicates that a significant portion of recent research has focused on defect segmentation using encoder-decoder-based architectures, which improves the results in defect segmentation. An interesting aspect of this category is the use of unlabelled data, where some works explore unsupervised learning techniques. These methods demonstrate the ability to learn from unlabelled data, making defect detection more accessible and flexible. By incorporating unlabelled data, the encoder-decoder models become more robust and adaptable to different scenarios. However, there is room to further improve accuracy and applicability for real-time applications.

5.3. Defect detection techniques

Defect detection in industrial vision is typically categorised into one-stage (e.g., SSD [81], YOLO [80]) and two-stage methods (e.g., Faster R-CNN [82]). For one-stage detection, Shafi et al. [176] uses SSD with MobileNet as the backbone for real-time steel defect detection. Zhang et al. [177] applies YOLO for detecting steel strip defects, leveraging anchor boxes and confidence maps for localisation. In two-stage approaches, Wang et al. [178] uses faster R-CNN for detecting small steel surface defects with 81% classification and 72% detection accuracy. Cha et al. [179] uses the same model for detecting concrete cracks and steel corrosion. Akhyar et al. [180] introduces a cascade R-CNN with a deformable region proposal network and random scaling to boost detection accuracy on steel surfaces. Recently, YOLO-based models have gained popularity due to their speed, with many studies enhancing performance using feature pyramids or integrating attention modules. This section highlights such developments.

5.3.1. YOLO with pyramid-based networks

This section focuses on defect detection using feature pyramid network with YOLO model. Liu et al. [40] proposes a YOLOv4 network for mapping small defective areas on the fabric where the spot is enclosed in the background. They use CLAHE to process the input image and spatial pyramid pooling structure instead of max pooling. It provides a real-time performance in industrial trials. However, limited resolution decrease the detection accuracy. Table 12 summarises papers that utilised the pyramid feature strategy for defect detection.

Han et al. [181] enhances YOLOv5 by integrating transformer modules with CSP-Darknet53, improving mAP by 3.1% for wood defects but struggling with fine cracks. An et al. [182] also works on wood defects by applying YOLOv8 with BiFormer, gaining 3.5% mAP at high computational cost. On other products, (e.g., LED, ceramic) Zhong et al. [183] proposes CM-YOLOv5 with multi-head self-attention, achieving a 3.8% mAP increase, while Chen et al. [184] uses BiFPN to reach 90.8% mAP. Chen et al. [185] uses improve YOLOv5 for ceramic defects, achieving 94.3% mAP through better multi-scale feature fusion. YOLOv8 later outperforms YOLOv5 in irregular surface environments. Zhao et al. [186] presents modified YOLOv8n for steel using dynamic coordinate attention and multi-scale blocks. Mei et al. [187] uses YOLOv8 for fabric defects using semantic fusion and enhanced attention, achieving 97.4% accuracy. Song et al. [188] further improves YOLOv8 with deformable convolution and C2F modules, boosting mAP by 6.9%. Despite these advances, most models still face challenges with detecting small, low-contrast defects in complex environments.

Sometimes complex defects are shown on the steel strip and ceramic production due to mechanical environmental factors. Detecting these

Table 12
Summary of contributions on defect detection using pyramid networks.

Reference	Object	Images/Dataset	Method	Accuracy	Function
Liu et al. [40]	Fabric	kaggle	YOLOV4	mAp: 98%	Pyramid
Han et al. [181]	Wood	3,606	Coordinate attention module, and transformer encoder in the YOLOv5	mAp: 84.2%	Pyramid
An et al. [182]	Wood	6,134	YOLOV8	mAp: increase in 5.8% from base model	Pyramid
Zhong et al. [183]	LED	128	YOLOv5 +Multi -Head Self -Attention mechanism	mAp: increase in 3.5% from base model	Pyramid
Chen et al. [184]	Mocro-LED	VOC2017	Multi-scale convolutional module + feature pyramid network +coordinate attention	mAp: 64.0%	Pyramid
Chen et al. [185]	Ceramic	1,000	Deep separable convolution +-Efficient Channel Attention in YOLOv5	mAp: 94.30%	Pyramid
Zhao et al. [186]	Steel	NEU-DET	Multi-Scale Adaptive Fusion with YOLOv8n	mAp: increase in 0.9% to base model	Pyramid
Mei et al. [187]	Fabric	2,116	YOLOv8n+attention+ feature pyramid network	mAp: 97.4%	Pyramid
Song et al. [188]	Steel	NEU-DET	Multi-directional optimisation based on the YOLOv8 algorithm	map: 84.3%	Pyramid
Du et al. [189]	Steel	NEU-DET	Adaptive focusing features (AFF-Net)	mAp: 83.5%	Pyramid
Peng et al. [190]	Metal	NEU-Det	ResNet50+ double attention FPN	mAp: 87.9%	Pyramid
Zhang et al. [191]	PCB	NEU-DET and PCB	CNN + attention mechanism+ feature pyramid network	mAP, NEU:78.64% and PCB: 91.95%	Pyramid
Zhao et al. [192]	Fabric, aluminium	Tianchi	Feature pyramid network (FPN) and path aggregation FPN	mAp increase in 1.4% to base model	Pyramid
Wan et al. [193]	Fabric, aluminium	13,000	Improved YOLOV5 based on spatial pyramid pooling	mAP: 87.6%	Pyramid

defects is essential for ensuring product quality and preventing economic losses in high-tech industries. Du et al. [189] proposes adaptive focusing features extraction method to better capture morphological and spatial changes in defects, combined with a feature pyramid network to enhance adaptability. Their method achieves 83.5% mAP on the NEU-DET dataset, outperforming the baseline by 8.2%. Similarly, Peng et al. [190] introduces a double attention feature pyramid network with deformable convolutions in the early ResNet50 stages, achieving 87.9% average precision for steel defect detection. Despite these advances, further improvements are needed to handle defects with random shapes. For surface defect detection on multi-object scenarios, Zhang et al. [191] proposes a lightweight CNN model with a coordinate attention backbone and a bidirectional weighted feature pyramid for feature extraction. While effective in detection, it struggles with complex defects. Zhao et al. [192] introduces an attention-based multi-scale feature fusion module that enhances defect-background discrimination using cross and self-enhanced attention but lacks precision across diverse defect types and demands high computational resources. To improve defect localisation, Wan et al. [193] develops an enhanced YOLOv5 model incorporating spatial pyramid pooling and attention modules. Evaluated on a colour printing fabric dataset, it achieves 87.6% mAP, outperforming baseline models.

The studies in this category focuses on improving results through the use of multi-scale features. YOLO-based models have improved in accuracy with pyramid features and attention, reaching up to 97.4% mAP. While YOLOv5 enhances speed over YOLOv4, both struggled with small or low-contrast defects. YOLOv8 offers better accuracy, but challenges persist in real-time deployment, generalisation, and robustness in complex environments. Despite ongoing advancements, balancing accuracy, speed, and reliability remains key for industrial use. However, challenges related to generalisation and real-time processing are the primary issues in this category.

5.3.2. YOLO with attention-based models

To enhance defect detection, researchers have integrated attention mechanisms with YOLO models, particularly through multi-scale feature fusion to improve small defect recognition. Studies by Pan et al.

[196], Li and Zhu [197], Mao et al. [198], Chen et al. [199], Cao et al. [200], and Li et al. [201] report notable gains in mAP, precision, recall, and F1-score across different surfaces such as ceramics, smart-phone glass, magnetic tiles, and other glass products. Despite these advancements, challenges remain in detecting defects with extreme aspect ratios and in reducing reliance on labelled data. Additionally, Lu et al. [202] employs a sliding window approach, which struggles with large defects exceeding the window size. Table 14 summarises these attention-based detection efforts.

Cui et al. [203] proposes an improved YOLOv7-based network for detecting small wood surface defects, introducing a cascade efficient layer attention module that improves mAP by 2.1% over the base model. Similarly, Zhang et al. [204] presents a lightweight model using a novel neck layer and squeeze-and-excitation attention to boost wood defect detection accuracy to 76.5%, improving mAP by 2.9% over YOLOv8, though it struggles with certain defects like knots with cracks. Lei et al. [205] introduces DGNet, a flexible model for battery defect detection, combining lightweight feature fusion and ShuffleNet backbone to achieve 91.8% mAP with reduced computation. Zhou and Wang [206] enhances YOLO with a triplet convolutional coordinate attention module to capture cross-channel features, outperforming baselines on NEU-DET and GC10-DET. Yuan et al. [207] proposes a model for multi-category steel defects using faster block, ghost convolutions, and channel attention in C2f, showing strong results on NEU-DET, though it struggles with detecting crazing defects due to labelling issues.

Ding and Yang [194] enhances YOLOv5 with constructive block attention and data augmentation to detect tiny defects on flat glass surfaces, achieving 98.36% and mAP 2.62% higher than the base model but does not address edge defects. Li and Yang [195] proposes a YOLOv5-based model for metal surface inspection using a hybrid attention module (e.g., channel, spatial) and dense connections, maintaining real-time performance with improved accuracy. Xu et al. [208] introduces a raster rotation imaging method for metal surfaces and integrates it into a YOLOv7-tiny model with multi-scale cross attention, improving mAP by 0.8% over the base model, though it still struggles

Table 13
Comparison of YOLO-based models and enhancements for industrial defect detection.

YOLO variants	Advantages	Limitations
YOLOv4 [40]	Introduces spatial pyramid pooling; Improves over YOLOv3 in mAP and speed	Struggles with small/low-contrast defects; Limited resolution and weak fine-grained features
YOLOv5 [194], [195]	Lightweight; Faster training and inference	Misses fine cracks and complex textures; Less robust in cluttered or noisy environments
YOLOv8 [186], [187]	Advanced backbones and BiFormer integration	High computational cost; Still struggles with generalisation and extreme defect variations
YOLO + Pyramid Networks (Table 12)	Better detection of defects at varying sizes	Increases latency and model size
YOLO with Attention Modules (Table 14)	Improves focus on defect regions; Enhances detection of tiny, subtle, or low-contrast flaws	Performance sensitive to annotation quality
YOLO + Coordinate Attention (Table 14)	Boosts both channel and spatial awareness; Enhances overall detection accuracy	Additional layers increase computation

Table 14
Summary of some contributions on defect detection using attention networks.

Reference	Object	Images/Dataset	Method	Accuracy	Function
Pan et al. [196]	Ceramic	–	YOLOV5+ ECA attention machismmm	mAp: increase in 4.5% from base model	Attention
Li and Zhu [197]	Fabric	NEU, GuangDong TianChi	Modified YOLOv5	mAp: 87.89%	Attention
Mao et al. [198]	Glass	>16,410	Dynamic head- YOLOv5+attention+crosslayer	mAp: 95%	Attention
Chen et al. [199]	Glass	>7,100	YOLOv5, MobileNetV3 and multi-scale feature fusion	mAp: increase in 3.9% from base model	Attention
Cao et al. [200]	Tiles	–	YOLOV5+attention machanism	Increase of 4.9% in AP	Attention
Li et al. [201]	Magnetic tiles	CALTECH-101, VOC 2007	Multi-resolution module with global and local attention	Acc: 99.50%	Attention
Lu et al. [202]	Tiles	–	YOLOv5 with Shufflenetv2+ path aggregation network	AP: 98%, F1-score: 97%, mAp 96.73%	Attention
Cui et al. [203]	Wood	–	Cascade center of gravity YOLOv7	mAp: increase in 1.2% from base model	Attention
Zhang et al. [204]	Wood	4000	Squeeze-and-excitation + attention mechanism in YOLO	mAp: increase in 2.98% compared to base model	Attention
Lei et al. [205]	Batteries	–	Adaptive lightweight backbone nwith feature fusion network	mAp: 91.8%	Attention
Zhou and Wang [206]	Steel	NEU-DET, GC10-DET	YOLO+ attention	mAP : 89.0%	Attention
Yuan et al. [207]	Steel	NEU-DET	Channel attention with YOLOv8	mAp : 75.8%	Attention
Ding and Yang [194]	Glass	–	YOLOv5+ Constructive Block Attention	mAp: 98.36%	Attention
Li and Yang [195]	Metal	–	Spatial +channel attention in YOLOV5	mAp: 98.9%	Attention
Xu et al. [208]	Metal	CSD-DET	YOLOv7-tiny + multi-Scale Cross Fusion Attention	mAp: 80%	Attention

with tiny defect detection. Feature enhancement is suggested to address this limitation.

Table 13 summarises the advantages and limitations of various YOLO variants, including those integrated with pyramid networks, attention modules for defect detection. From the table, it is revealed that, integrating attention mechanisms with YOLO models has led to improved performance in defect detection, particularly for small and subtle defects across surfaces, such as ceramics, glass, and metals. Enhancements such as channel and spatial attention, coordinate attention, and multi-scale fusion have significantly boosted metrics such as mAP and precision. Lightweight versions (e.g., DGNNet, YOLOv7-tiny) also offer real-time potential with reduced computational load. However, these models still face limitations in detecting defects with extreme aspect ratios, irregular shapes, or poor annotations. Sliding window methods struggle with larger defects, and some attention-augmented models underperform on specific defect types (e.g., knot with crack or

crazing). Despite high accuracy in controlled settings, generalisability and robustness across diverse real-world scenarios remain ongoing challenges.

5.4. Unsupervised, semi-supervised and other learning for defect detection

This section highlights studies on unsupervised model (e.g., GAN-based, diffusion), semi-supervised, weakly supervised, and self-supervised (e.g., contrastive learning) models for defect detection. All the methods are discussed in the following subsection.

5.4.1. GAN-based models

Zhang et al. [131] introduces a multi-scale progressive GAN with self-attention and non-leaking data augmentation to address limited sample availability. The synthetic images generated enhance the convergence speed and accuracy of the defect detection model, which

Table 15

Summary of some contributions on defect detection using unsupervised learning. Note: GAN: Generative Adversarial Network, AE: Auto Encoder.

Reference	Object	Images/ Dataset	Method	Accuracy	Function	Category
Zhang et al. [131]	Steel	GC10, NEU-DET	Multi-scale progressive GAN networks	mAp: 95%	GAN	Detection
Ou et al. [209]	Ceramic	1353	StyleGAN with few shot	mAp: increase to 1.5% compared to YOLOv5	GAN	Detection
Xie et al. [210]	Multi-object	–	High-quality Matching Transfer GAN	–	GAN	Detection
Yang et al. [211]	Multi-object	–	Frequency shift-convolutional AE	mAp: 95.19%	AE	Detection

is further evaluated using YOLOv5. Ou et al. [209] addresses the challenge of data labelling and collection by proposing a StyleGAN3-based data augmentation method to generate ceramic defect samples, reducing training time and improving detection accuracy. Xie et al. [210] introduces HMTGAN, a CycleGAN–StyleGAN hybrid model, using latent space sampling and a novel loss function to generate high-quality defect samples, achieving strong results on crack datasets. Yang et al. [211] presents an unsupervised approach using a frequency shift convolutional auto-encoder trained on defect-free samples, addressing low sample size and reducing manual labelling effort. Table 15 presents a summary of selected papers in the field of GAN networks for defect detection. The studies conduct in this category have demonstrated promising results, particularly in the context of unsupervised and weakly supervised approaches. These approaches offer valuable alternatives for detecting defects without relying heavily on labelled data.

5.4.2. Diffusion-based models

Diffusion models are emerging as a powerful tools in industrial defect detection and augmentation, addressing data scarcity and improving robustness. These generative models learn to synthesise realistic images by denoising random noise through a learned reverse process, typically trained in an unsupervised fashion. MED-AGNeT [212] combines a conditional diffusion module with attention-based detection to improve accuracy on textile defect detection. Tang et al. [213] enhances unsupervised fabric defect localisation using timestep-adaptive denoising. In steel defect detection, Tai et al. [214] applies stable diffusion with token and domain adaptation to improve accuracy. Table 16 presents a summary of selected papers in the field of diffusion networks for defect detection. It is showed in the table that Latent Diffusion Models (LDMs) and DiffAD are used to reconstruct normal patterns and isolate anomalies with pixel-level precision, as seen in both steel [215] and general industrial surface defect detection [216]. Hong et al. [217] uses YOLO-DSC with DDIM-based augmentation for efficient real-time detection. The overall study shows that diffusion models, whether used for data generation or reconstruction can substantially enhance unsupervised and semi-supervised defect detection pipelines.

5.4.3. Semi-supervised learning

Semi-supervised learning has proven effective for defect detection in data-scarce industrial settings. Table 17 presents a summary of selected papers in this field. For example, Lei et al. [218] introduces a hybrid U-Net with CSPNet and CBAM, augmented with synthetic data through Poisson noise, achieving 98.22% mAP on GC10-DET. This approach demonstrates strong generalisation but may struggle in uncontrolled real-world conditions. Similarly, Song et al. [219] introduces MSD Net, combining CycleGAN-based domain transfer with pseudo-labelling and multi-head self-attention, showing a 4.8% improvement over supervised baselines. However, its reliance on domain migration and attention modules introduces complexity. Another method, DifferNet [220] learns from only normal samples but suffers from noisy localisation on large defects. Cao et al. [221] presents a knowledge distillation method with abnormal-aware feature enhancement, though

it depends heavily on the quality of the teacher model. Lastly, Shi et al. [222] proposes a lightweight DeepLabv3+ and MobileNetV2-based model enhanced with dynamic thresholding and cross-pseudo supervision, achieving over 86% mIoU, face limitations in adverse conditions (e.g., motion blur).

5.4.4. Self-supervised learning

Self-supervised learning (SSL) is increasingly recognised as a robust method for industrial defect detection, particularly where labelled data is scarce. SSL enables models to learn from unlabelled data via pretext tasks, such as contrastive learning which constructs a feature space by using similar instances and repelling dissimilar ones using techniques such as SimCLR [223] with strong data augmentations and encoders, such as ResNet and ConvNet. One of the foundational methods, CutPaste by Li et al. [224], introduces synthetic anomaly generation through patch replacement, enabling robust training on only normal samples and achieving 95.2% AUC on MVTec AD. However, the synthetic patches may not fully mimic real defect patterns. Building upon this, Lv et al. [225] enhances spatial awareness using ECA-Net and CAM-loss, boosting AUC to 0.946, although manual tuning is required. Table 18 presents a summary of the papers in the field of SSL networks for defect detection. For example, Zabin et al. [226] uses SimCLR framework with lightweight encoders, attaining 97.78% accuracy on NEU while reducing computational complexity, however, sensitive to augmentations and batch sizes. In other works, Hu et al. [227] applies SimSiam and earth mover's distance for multi-target detection, and Wu et al. [228] fuses contrastive and inpainting objectives, achieving strong generalisation across five datasets. Despite its robustness, it suffers from performance drops on small defects and handcrafted pretext tasks. Xu et al. [229] designs SEDD, a lightweight segmentation model with homographic pseudo-labelling and spatial-channel attention, delivering 98.4% AUC, suitable for real-time applications but limited under texture-rich scenarios. Wang et al. [230] deploys advanced few-shot learning via TSMMSIS, a self-supervised teacher–student framework using min–max instance similarity, achieving 95.5% accuracy on NEU but remaining sensitive to network depth and hyperparameter settings. Huang et al. [231] proposes ACViT-SCL, integrating meta-learning with SimCLR and adaptive cross-transformers, performing well in low-shot regimes.

Although SSL enhances representation learning in label-scarce settings, challenges remain in model calibration, sensitivity to augmentations, and real-world deployment efficiency. Future work may benefit from hybrid SSL-few-shot-transformer approaches to balance generalisability, performance, and real-time feasibility.

5.4.5. Weekly-supervised learning

There are also weekly supervised learning (WSL) models that are used when there is a scarcity of enough labelled data to train the model. Table 19 presents a summary of the contribution of WSL in defect inspection. For example, Wu et al. [232] presents a WSL model by utilising image-level annotations, a siamese network is utilised to close the gap between image-level and pixel-level supervision. To improve performance without increasing computing complexity, the

Table 16
Summary of diffusion-based contributions for defect detection.

Reference	Object	Images/ Dataset	Method	Accuracy	Function	Category
[212]	Fabric	Custom	MEDiffusion	TPR: improve to 1.88% compare to others	Diffusion	Classification and Detection
[213]	Fabric	In-house	Diffusion with Adaptive Timestep	F1-score: improve to 5.42%	Diffusion	Detection
[214]	Steel	NEU, CCBSD	StableSDG (Stable Diffusion with Token Adaptation)	Accuracy: improve to 10%	Diffusion	Classification and Detection
[215]	Steel	NEU-seg	Stable Diffusion with DeepLabV3+	–	Diffusion	Segmentation
[217]	Steel	NEU-DET	YOLO-DSC	mAP: 80%	Diffusion	Detection
[216]	Multi-object	MVTec-AD	DiffAD	AUROC: 98.3%	Diffusion	Detection

Table 17
Summary of semi-supervised learning models for industrial defect detection.

Reference	Object type	Dataset(s)	Key Features/Notes	Accuracy	Category
[218]	Steel	GC10-DET	Hybrid U-Net with CBAM and Poisson noise for augmentation	mAP: 98.22%	Segmentation
[219]	Steel	Steel Benchmark, VOC 2007	CycleGAN for domain adaptation, pseudo-labelling, multi-head self-attention	mAP@0.5: 96.1%	Detection
[220]	Multi-object	MVTec AD, Magnetic Tile	Normalising flow model trained on normal samples, unsupervised localisation	AUROC: up to 97.7%	Detection
[221]	Plastic	Custom (10,950 patches)	Teacher–student distillation with abnormal-aware feature enhancement	AUROC: 95%, F1-score: 92%	Detection
[222]	Steel	KolektorSDD	Lightweight DeepLabv3+ with MobileNetV2 and dynamic thresholding	mIoU: up to 86.32%	Segmentation

approach made use of log summation exponential global pooling modules and max pooling-based non local attention. Xu et al. [233] uses VGG-based CNN with residual attention to detect defects in KolektorSDD using minimal labels. Luo et al. [234] proposed a memory-attended saliency framework combining FPN and attention, achieving strong results across DAGM and Severstal datasets. DefectGAN [235], a CycleGAN-based approach, achieved 81.05% IoU without pixel-level ground truth. Huang et al. [236] introduces an attention-enhanced nested U-Net++ reaching 100% AUC, though limited by complexity and synthetic data reliance. While WSL models reduce annotation costs and adapt well to various defects, their performance can be hindered by generalisability, deployment readiness, and reliance on synthetic or adversarial components.

5.4.6. Discussion

In summary, compared with pyramid networks, attention based networks have advantages in defect information acquisition. However, for supervised techniques, similar to segmentation networks, detection networks also demand a substantial amount of labelled data, with pixel-level annotations. This often requires a significant effort and cost for labelling. However, defect detection methods based on semi-supervised and self-supervised methods are relatively more flexible and can take advantage of limited or incomplete data labelling. Therefore, in practical applications, an appropriate detection method can be selected according to the specific data situation and resource constraints.

6. Other networks

Aside from YOLO-based models with feature pyramids and attention mechanisms, other models based on transformer architectures, LLM and foundation models are also used for defect detection and segmentation.

6.1. Transformer-based models

Transformer-based models have revolutionised industrial defect detection by effectively capturing long-range dependencies and global contextual features. After the development of attention mechanisms, transformer-based models such as Vision Transformer (ViT), Swin Transformer, and DETR variants have demonstrated strong performance in segmentation, classification, and detection tasks, especially in steel, PCB, and textile defect inspection [238–240]. Table 20 presents a summary of the papers used transformer-based network for defect detection. Su et al. [238] introduces RT-DETR with CBAM for printed fabrics, while MS-DETR [240] achieves high performance in real-time PCB defect detection. Models such as RTCN [241], LSwin [243], CGTD-Net [245], and LMViT [247] further pushed boundaries in steel defect segmentation and classification. Chen et al. [258] suggests a model named EEE-Net which is an effective edge enhancement network that also uses transformer blocks and sequence reduction to improve detection accuracy, it incorporates an edge encoder, decoder and edge information fusion module. Experimental results on glass and mobile phone screen demonstrate the model's superior performance and suitability for real-time industrial applications.

Gao et al. [259] also implements a framework for surface defect detection utilising the transformer architecture. Their method employs patch embedding in the VSwin transformer to divide the input image into overlapping patches. These patches are able to extract more abstract data from the original input pixels. Afterwards, features from several patches are taken out and used to defect detection modules. When trained on small-scale datasets, their methodology outperforms transfer learning techniques. However, these models often face practical limitations including high memory usage, long inference times, and the need for large annotated datasets, making real-time deployment difficult. Su et al. [242] implements a model named PMSA-DyTr based on semantic aligners, dynamic encoder provides accurate results, rely on intricate modules that increase system complexity and

Table 18

Summary of self-supervised learning models for industrial defect detection.

Reference	Object	Dataset	Method	Accuracy	Category
[224]	Multi-object	MVTec AD	Patch-based SSL using synthetic cut-paste patches, label-free training	Acc: 95.2% detection and 96.0% localisation	Detection and localisation
[225]	Multi-object	MVTec AD	Enhanced spatial focus using ECA-Net and CAM-loss	AUC: 94%	Detection
[226]	Steel	NEU, Severstal, GC10	Contrastive learning with SimCLR/NNCLR + lightweight CNN encoder	Acc: 97.78%	Detection
[228]	Multi-object	NEU-DET, RSDDs, KolektorSDD2	Fusion of contrastive and restorative tasks (GC/LC contrast + inpainting)	SOTA on 5 datasets	Detection and classification
[227]	Multi-object	NEU-DET, GC10-DET	SimSiam-based learning using Earth Mover's Distance (EMD)	mAP improved to 6.8%	Detection
[229]	Steel	Crack, Kolektor	Lightweight encoder with homographic pseudo-labelling and attention	AUC: 98.4%, AP: 74.84%	Segmentation
[230]	Steel	NEU, X-SDD, GC10	Few-shot and Teacher-Student framework with instance similarity discrimination	Acc: 95.5% in NEU	Detection
[231]	Steel	NEU, PKU-PCB, CLD	SimCLR backbone with adaptive cross-transformers and Few-shot	Acc: 72.26% (10-shot)	Detection
[237]	Multi-object	MVTec AD	ViT with deterministic mosaic inpainting and saliency masks	AUROC: 92%	Detection

Table 19

Summary of weakly supervised learning models for industrial defect detection.

Reference	Object	Dataset	Method	Accuracy	Category
[233]	Multi-object	KolektorSDD	VGG with residual attention	–	Segmentation
[234]	Multi-object	DAGM, KolektorSDD, Severstal	Saliency heatmap with memory fusion	–	Segmentation
[236]	Multi-object	MVTec AD, MT, CT, DAGM	Nested U-Net++ with attention	AUC: 99.9% (MVTec), 99.1% (DAGM), 73.6% (MT)	Segmentation
[235]	Multi-object	CCSD-NL	Image translation and CycleGAN-based	IoU: 81.05%	Segmentation

over-detection risks. Likewise, JDCBL [248] incorporates a contextual transformer with contrastive loss for fabric inspection, but suffers from slow inference and manual tuning. To overcome these constraints, few-shot learning methods have been explored as lightweight, data-efficient alternatives. MAPNet [239] and CPANet [249] show strong segmentation capabilities on small datasets, while EidNet [250] and DSRF [252] are tailored for classification and detection in low-sample environments. MFANet [251] effectively bridges coarse and fine defect granularity but can be computationally intensive. Although few-shot models reduce reliance on labelled data, they sometimes struggle with subtle or ambiguous defects. Ultimately, integrating transformer architectures with few-shot or self-supervised learning represents a promising path for building robust, scalable, and generalisable defect detection systems that suit the diverse demands of real-world industrial applications.

Foundation models (FMs), including SAM, CLIP, and GPT, and Large Language Models (LLMs) have emerged as promising tools for industrial defect detection due to their strong generalisation capabilities and ability to operate in zero and few-shot learning scenarios [260]. These models are pretrained on extensive datasets and can be fine-tuned for specific tasks with minimal labelled data, making them well-suited for data-scarce environments. Table 21 presents a summary of the papers in this field for defect detection. For example, Jiang et al. [254] develops PA-SAM, an improved variant of SAM for efficient fine-tuning, and another module (e.g., IPEG) for automated prompt generation, achieving notable improvements on datasets, such as SD-Saliency-900 and Tire-Seg. LLMs are also gaining attraction for their ability to integrate vision-language reasoning and support multi-modal interactions. Zhang et al. [255] leverages ResNet-18 and fuzzy inference with LLMs to generate interpretable defect explanations. Similarly, AnomalyGPT [256] enables open-set detection and multi-turn dialogue

for defect inspection, while Boonmee et al. [257] utilises GPT-4 for diagnosing industrial faults through iterative prompting. Despite their strengths, FMs often demand substantial memory and computational resources, and they struggle to adapt flexibly to domain-specific features without further tuning.

7. Highlights of the study

The methodology and accuracy of the studying research papers on DL-based methods for defect detection in industrial applications are described in the preceding section. This section outlines the statistical analysis of the defect detection techniques. According to the taxonomy described in Fig. 1, the entire literature is broadly categorised into two groups: semi-automated and automated defect detection methods. Fig. 8 compares the number of studies focusing on semi-automated and automated defect detection methods. We observe that automated methods based on DL approaches dominate, with 141 papers, while semi-automated methods have only 27 papers. Both supervised and unsupervised methods are investigated in the DL-based automated methods; shown in Fig. 9. It shows that supervised learning dominates the field with nearly 101 studies, followed by unsupervised learning (25 papers). In contrast, self-supervised, semi-supervised, and weakly-supervised approaches are significantly less explored, with fewer than 20 papers each. This indicates a growing research opportunity in alternative learning methods to address challenges such as data scarcity and annotation costs.

Later, supervised techniques are further subdivided into tasks related to classification and localisation, where localisation is further subdivided into segmentation and detection. Fig. 10 illustrates the distribution of defect classification, segmentation and detection functions used to recognise defects in industrial applications. The results indicate

Table 20

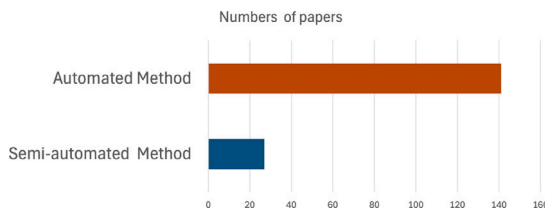
Comprehensive summary of Transformer and Few-shot model-based defect detection contributions. Note: Ap: Average precision.

Reference	Object	Dataset	Method	Accuracy	Function	Category
[238]	Fabric	9,828	RT-DETR with CBAM attention	AP: 58.8%	Transformer	Detection
[239]	Steel	FSSD-12	Multi-scale adaptive prototype transformer	mIoU: 66.5% (1-shot)	Few-shot	Segmentation
[240]	PCB	3,283	MS-DETR with Hilo Attention	mAP: 96.9%, APs: 68.3%	Transformer	Detection
[241]	Steel	NEU-DET	Swin transformer with deformable convolutional block	mAP $\uparrow$ 3.8%	Swin Transformer	Detection
[242]	Steel	NEU-DET	Long short-term self-attention embedded with local convolution and semantic-aligned transformer	mAP: 81.2%	Semantic Transformer	Detection
[243]	Steel	NEU-DET, GC10	LSwin Transformer	mAP@0.5: 81.2%	Transformer Hybrid	Detection
[244]	Steel	NEU-DET	ViT	Accuracy: 96.39%	Vision Transformer	Classification
[245]	Steel	NEU-DET, etc.	Channel-Wise Global Transformer and Swin transformer	mIoU: 75.16%	Dual Transformer	Segmentation
[246]	Fabric	ZJU, HKU	CNN with Swin transformer	F-measure: 75.33%	Hybrid	Segmentation
[247]	Steel	NEU-DET	Memory-Augmented ViT	Acc: 97.22%	Transformer	Classification
[248]	Fabric	Tianchi	Joint Dynamic Contexts and Balanced Learning with contextual transformer	mAP@50: 39.3%	Contextual Transformer	Detection
[249]	Steel	FSSD-12	Cross Position Aggregation Network with ResNet50 backbone	—	Few-shot	Segmentation
[250]	Steel	FSC-20	EidNet (ResNet18 and extending interclass distance)	Acc: 95.89% (10-shot)	Few-shot	Classification
[251]	Tubes	CGFSDS-9	Multi-feature guidance strategy and triplet prototype module	mIoU: 75.45%	Few-shot	Segmentation
[252]	PCB	DeepPCB	Selective feature enhancement in foreground information and channel regulation transformation in detection	mAP@50: 80.2% (10-shot)	Few-shot	Detection
[253]	Steel defects	NEU-DET (1800)	Simple sparse sliding-window attention, which localises self attention for each pixel to its near neighbours.	mAP: 83.7%	Sparse Attention Transformer	Detection

Table 21

Summary of foundation models and LLM-based contributions for industrial defect detection.

Reference	Object	Dataset	Method	Accuracy	Category
[254]	Multi-object	SD-Saliency-900, Tire-Seg	Lightweight multi-scale partial convolution aggregation to enhance the image encoder, and another module, IPEG, which autonomously creates high-quality prompt embeddings for directing mask segmentation	—	Segmentation
[255]	Steel	Kaggle	Hybrid XAI with fuzzy LLM interpretation and Genetic algorithm to optimise the parameters	Acc: 81.03%	Detection
[256]	Multi-object	MVTec-AD, VisA	AnomalyGPT (ImageBind and prompt learner) with open set detection	AUC: 95.3%	Detection
[257]	Multi-object	Sensor data	GPT-4 with multi-round prompting and structured consultation	Acc: 95%	Detection

**Fig. 8.** Distribution of semi-automated and automated method for defect detection in industrial applications.

that the segmentation and detection techniques together are the most commonly employed, with 101 articles (44 on segmentation and 57 on detection) focusing on highlighting the specific pixels corresponding

to the defects while 34 studies focusing on identifying the presence or absence of defects on the surface.

In addition, we categorised DL-based models into several aspects: CNN-based, transfer learning, and hybrid models for defect classification and encoder–decoder, pyramid networks, YOLO-based, attention-based models for defect localisation. Our analysis shown in Fig. 11 reveals that CNN-based models are most prominent in defect classification, representing 37% of studies. In segmentation, encoder–decoder models dominate with 39%, while YOLO models with attention mechanisms lead in detection tasks. This highlights a trend in recent research towards attention-based architectures for defect detection, owing to their ability to extract features at multiple levels and focus on important features. Moreover, Fig. 12 shows the evolution of publication trends in defect detection, segmentation and classification methods from 2012 to 2025.

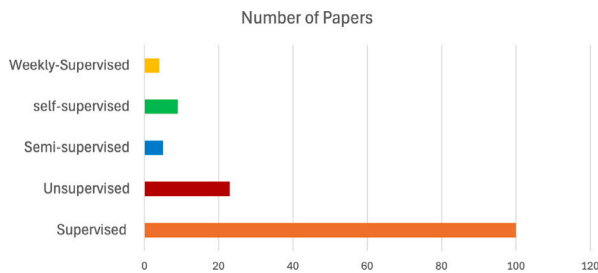


Fig. 9. Distribution of supervised and unsupervised models for defect detection in industrial applications.

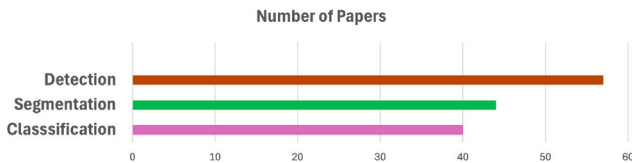


Fig. 10. Distribution of detection, segmentation and classification functions in defect detection in industrial applications.

From Fig. 12, it is revealed that initially classical ML and image processing techniques were predominant, particularly between 2012 and 2016. However, their usage gradually decreases as DL-based methods became more prominent. From 2017 onward, there is a clear indication in the adoption of CNN and transfer learning, reflecting the field's shift towards data-driven approaches. Around 2019, encoder-decoder architectures started gaining popularity, primarily for segmentation tasks. Attention-based and pyramid networks also have a gradual rise, indicating a growing focus on enhancing feature extraction and multi-scale representation. More recently, GAN, transformer-based architectures, and semi/self/weakly supervised learning methods have gained momentum. This trend highlights a movement towards models that can either generate synthetic data, work with limited labelled samples or handle more complex visual patterns. The growing trend of these modern techniques reflects continuous innovation aimed at improving the efficiency and accuracy of defect detection in various industrial sectors.

There are numerous datasets used across various domains, including steel, fabric, metal, glass, and other industrial materials. The models, datasets, and image sizes used in each work are reported in Tables 9 to 21. Among these, several works on defect classification provide the size of the datasets, while others mention only the dataset name. The works which specify the size of the datasets, we created a bubble chart illustrating how different models perform with varying dataset sizes. Fig. 13 shows the relationship between dataset size and accuracy across different model types for defect classification. The chart highlights that high accuracy is often achieved with larger datasets, and transfer learning performs well even on medium-sized datasets, while hybrid models also demonstrate competitive performance. Furthermore, for defect segmentation and defect detection tasks, a variety of datasets are used. The four most prominent datasets are DAGM, NEU-DET, MvTec, and KolektorSDD. Other datasets are also utilised across many studies. Model performance is commonly evaluated using metrics such as mAP and mIoU. The methods and their reported accuracies are summarised in Tables 9 to 21. Additionally, Fig. 13 visualises the dataset-accuracy relationship across methods for these four datasets, highlighting several key insights. DAGM and NEU-DET show a wider spread in performance, especially for ED and attention-based models, suggesting variability in model implementation or dataset complexity. MvTec shows consistently high accuracy, particularly with SSL and LLM models, indicating that these approaches are highly effective for this dataset. KolektorSDD, on the other hand, shows a sharp contrast

while SSL and GAN models achieve very high accuracy, pyramid-based methods perform poorly on this dataset. This comparison underscores the importance of model selection based on dataset type and defect complexity.

8. Open challenges and future research directions

Automated inspection systems are becoming more and more crucial to quality assessment in the industry sector. Some researchers published review articles [27–30] on defect detection and classification in industrial products, but there are still many challenges left to tackle in this domain. This survey provides a comprehensive study of defect detection and classification methodology using image processing, ML and DL methods and describes how they are applied to the different products. This section presents the identified open challenges that are categorised as follows: (1) data pre-processing, (2) deep learning model, and (3) model reliability and trustworthiness. Fig. 14 represents a schematic diagram of future research direction visualising all critical components where challenges arise in the model for defect detection. In addition, some findings are investigated that should be considered in the future.

8.1. Data pre-processing in industrial settings

This section analyses the key challenges in collecting the dataset for defect inspection, as it is a necessary step for all DCNN models. Databases come with a series of challenges such as data cleaning, data labelling, data privacy and data sharing.

- **Data Cleaning:** The first challenge comes when collecting images from an industry, there are limitations in capturing the images such as non-uniform illumination and camera noise. Thus, this image can run into some issues, such as blurred images, defects that are not clearly visible or noise. So, this kind of image needs to be preprocessed before feeding them into the model. Cleaning the noise increases the data quality and makes the model more robust. Then data need to be resized as per model choice.
- **Data Labelling:** The next challenge is data labelling, which is also crucial to the preparation of the dataset for the model, and therefore necessitates a lot of human work. Automated defect detection faces significant challenges due to data privacy, particularly in some domains where the industry keeps the data confidential. Moreover, some researchers do not like to share data publicly, which makes it difficult for other researchers to collect the dataset and train their models.

Nahar et al. [261], develops a standardised data capturing policy for automated defect detection in trading cards. In the pre-processing stage, data undergoes noise reduction via a bilateral filter, ensuring clean inputs. Following this, an algorithm was implemented to crop portions of specific sizes from the cards, thereby establishing a database with uniform dimensions. In another research [229], a custom dataset for steel defect detection is created by systematically cropping data into predetermined patch sizes, each representing different defect types. However, in both cases, the datasets are initially labelled manually. In another study [262], a ssl is proposed as a solution to the challenge of dataset labelling.

8.2. Deep learning models: Refining network selection and parameter tuning

This section addresses the challenge of choosing a network for defect detection and classification. Three categories of defect detection methods are discussed in the previous section: manual, semi-automated and automated. Semi-automated methods use image processing techniques (e.g., filter-based, edge-based), which can solve the problem of manual inspection such as no need of experimental setup, including subjectivity, high labour expenses, and mistakes. Classical ML

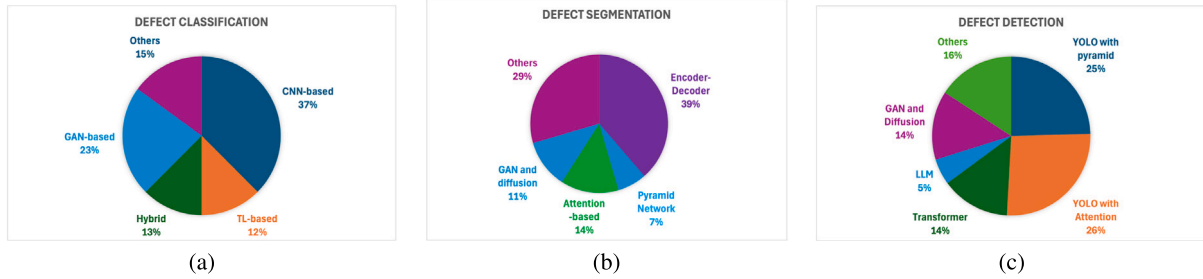


Fig. 11. Percentage of deep learning approaches for defect detection in industrial application: (a) Classification, (b) Segmentation and (c) Detection.

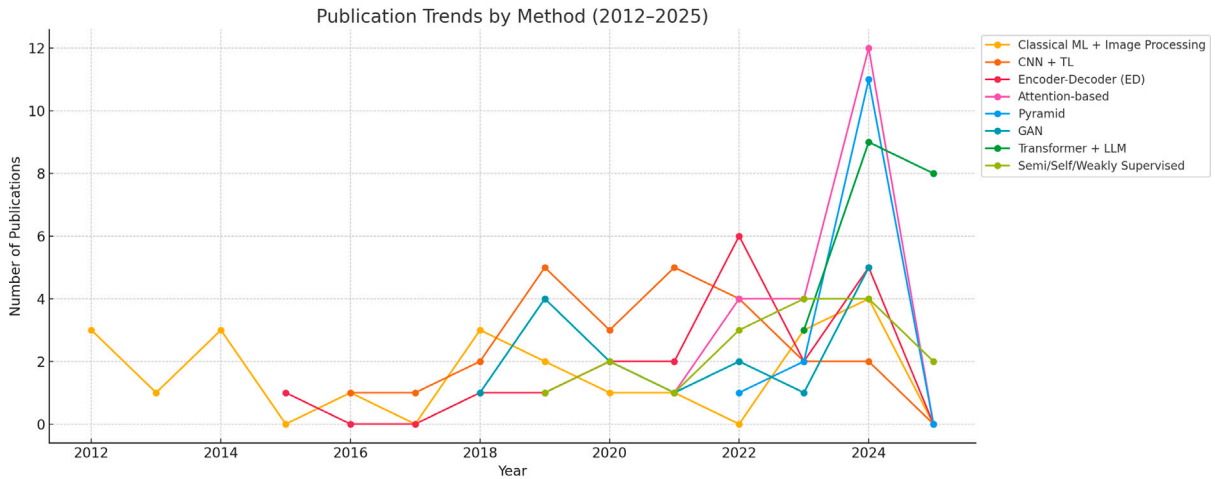


Fig. 12. Publication trends in classical machine and modern learning methods in product defect detection.

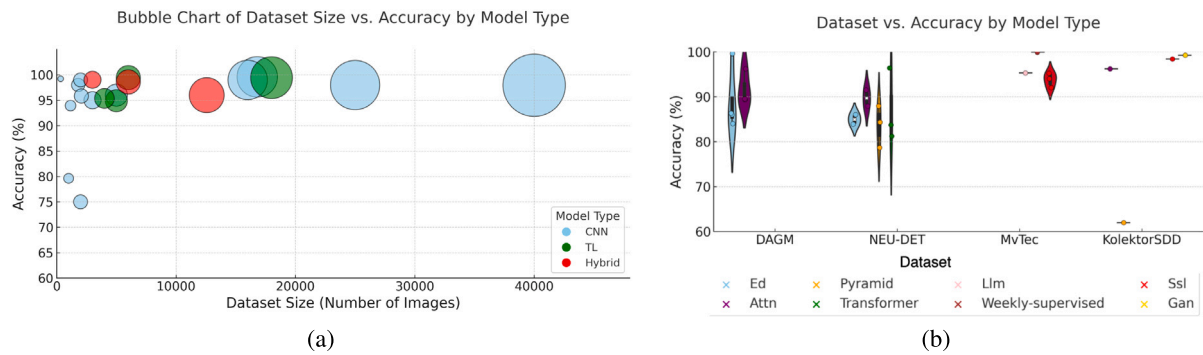


Fig. 13. Accuracy on different datasets on various applications: (a) Classification (b) Segmentation and Detection.

based semi-automated also solve these challenges but face other issues (e.g., poor feature extraction). Automated defect detection methods offer significant detection accuracy and cost reduction. They are effective in detecting defects in diverse patterns but they also have drawbacks, including the requirement for large amounts of labelled data, imbalanced datasets and hyperparameter tuning described below. Hence, by taking these three contexts the following research questions (RQs) require attention:

- RQ1: How can we employ deep learning models effectively when dealing with small datasets?
- RQ2: Which hyperparameters need to be tuned for optimal performance in a specific task?

8.2.1. RQ1: Addressing the problem of small dataset in dl-based models

Automated defect detection in industrial settings encounters a fundamental challenge: the scarcity of labelled defective data compared

to the abundance of correctly labelled samples. This imbalance can skew model training and hinder its ability to generalise effectively. To mitigate this, various techniques such as regularisation, early stopping, and data augmentation, including shifting, flipping, and zooming, are applied to bolster model robustness and adaptability to new data [23, 38,261,263].

8.2.2. RQ2: Hyperparameter selection for models optimisation

The selection of an appropriate network architecture and fine-tuning hyperparameters including adjustments to parameters like the number of layers, stride, filter size, batch size, selecting the appropriate optimiser and learning rate are crucial steps influenced by factors like dataset size and desired detection accuracy. Determining the number of neurons and activation functions further adds to the complexity of network design. Overall, achieving generalisable model accuracy requires careful consideration and fine-tuning of these parameters [14,

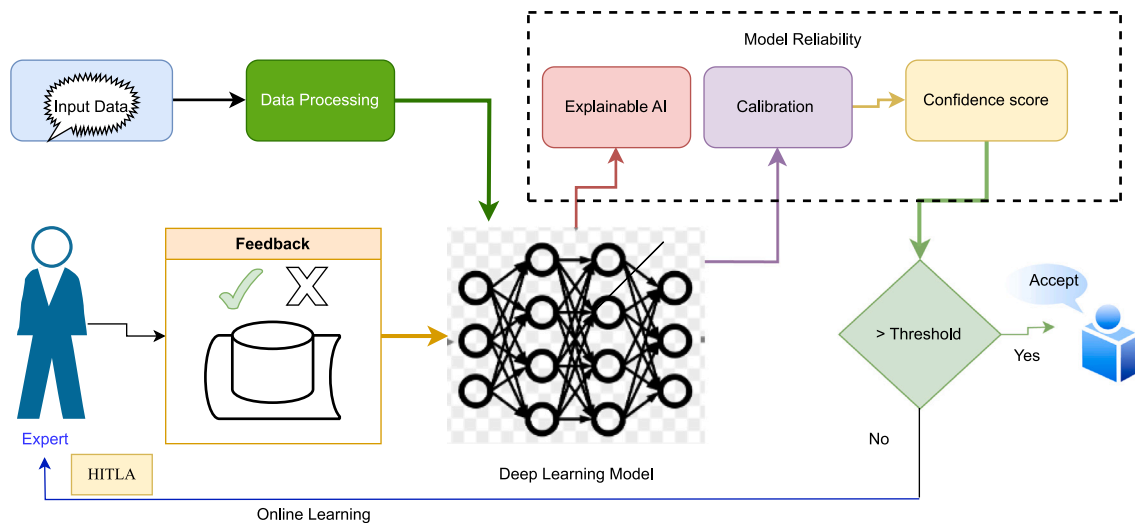


Fig. 14. A schematic diagram of the proposed research direction. Note: TL: Transfer learning, HITLA: Human-in-Loop Analytics.

[139]. Defect localisation adds another layer of complexity, often addressed by specialised semantic networks such as Fully Convolutional Network [157] and U-Net [74]. However, these methods entail labour-intensive pixel-wise annotation, prompting the development of unsupervised models (e.g., GAN [264]) which alleviate the annotation burden at the expense of slightly lower accuracy compared to supervised approaches. These strategies collectively strive to optimise defect detection systems for real-world industrial applications.

8.3. Defect detection models: Navigating reliability and explainability

This section highlights a great challenge in defect detection model. Although DL has emerged as a popular field for effective defect detection in various industries, recent works have pointed out that the outputs provided by these models are not well-calibrated, limiting their use in critical decision scenario. Another great Challenge in CNN's network is explainability because they work as a black box, which is hard to understand how the decisions are being made. Moreover, after training a DL model, when it is applied to a real-world task, it is important to identify that no model is fault less, and there may be occurrences where its predictions may be uncertain. To refine and assess DL models, human expertise and knowledge are integrated into the DCNN. In this perspective, decisions can be made based on feedback from experts or by retraining the model. Henceforth, the scenario formulate some research questions.

- RQ3: How can trust be ensured in the decisions made by the black-box model (e.g., DL)?
- RQ4: How much data is required to retrain the model?

8.3.1. RQ3: Reliability challenges and solutions for DL-based models

This subsection describes the reliability of deep learning models in industrial defect detection, focusing on explainability, uncertainty estimation, and calibration. Together, these three aspects determine how trustworthy and robust a model is when deployed in real-time decision-making environments.

(a) *Models reliability through explainable AI.* This section delves into the challenges of decision-making and ensuring model reliability in real-time applications, particularly in CNNs. A major hurdle lies in the explainability of CNNs, given their black box nature, which makes understanding decision-making processes challenging. While CNN layers identify patterns used in decision-making, it is often unclear which patterns or features are prioritised and why. This lack of transparency poses significant concerns, especially in critical domains like medicine

and autonomous vehicles, where understanding decision-making is paramount. To tackle this, researchers focus on Explainable AI (XAI), aiming to enhance reliability, trustworthiness, and transparency of AI systems. Various XAI methods, such as saliency map [265,266], Grad-cam explored to elucidate CNN decisions. While XAI methods enhance transparency, maintaining model confidence remains crucial for accurate decision-making. Other papers [267,268] also offers a review of XAI approaches in different applications.

(b) *Uncertainty estimation of DL-based models:* Another challenge in the performance evaluation of DL based models is uncertainty quantification (UQ) as existing defect detection methods are implemented based on clean, good and uncontaminated training data. The ability for UQ using current AI-based methods is not only limited in terms of computational resources but it also requires changes to topology and optimisation processes, as well as multiple performances to monitor model instabilities. To address this challenges Abbaszadeh Shahri et al. [269] proposes a novel solution, Automated Random Deactivating Connective Weights (ARDCW), without changing the optimisation processes. In a single feed-forward pass ARDCW can effectively evaluate the uncertainty and it does not require changes in the optimisation process, can be applied to already trained topology in a way that outperforms other models. Yu et al. [137] addresses image uncertainty due to ambient light and label uncertainty due to human errors in PCB surface inspection. The proposed learning model overcomes this limitation by using a symmetric residual filter and probabilistic classification. It is further improved by combining a joint probabilistic inference engine. These models performed well in both self-collected and public datasets. Their experimental result showed that it can detect defects in complex patterns also. Therefore, there are still challenges that need to be addressed. Future work in this research field should be taken into account by combining the models in such a way so that to yield a more lightweight model while ensuring high detection accuracy [146,270,271].

(c) *Calibration techniques for enhancing DNN confidence:* In the literature, it is shown that DNNs consistently shows excellent performance across a wide variety of tasks. However, they often suffer a lack of proper calibration. In real-time decision-making applications, such as quality inspection in industry, the confidence levels of deep networks are essential for instilling reliability in the system. To address this calibration challenge, various probabilistic and measurement-based methods have been proposed, including Temperature Scaling, Beta Calibration, Platt Scaling, Isotonic Regression, Matrix Scaling, and Vector Scaling [272–274]. These methods allow the calibration of DL models,

yielding confidence scores for test data. Decisions can be made by selecting the highest probability from this confidence score. A well-calibrated model can make a great impact on decision-making in defect classification and detection problems [263]. Lei et al. [275] proposes a surface defect detection method, where they implemented an adaptive convolution confidence sieve learning method under process uncertainty. They combine adaptive convolution, edge-separated wavelet transform, and confidence sieve learning, which enhance accuracy and robustness. Their experimental result shows that these methods perform well in industrial datasets.

8.3.2. RQ4: Integrating HITLA in DL-based models refinement

After training a DL model, it is crucial to acknowledge that no model is perfect, and there may be instances where its predictions are uncertain. HITLA refines and assesses DL-based models outputs by integrating human expertise and knowledge [276,277]. In HITLA, the inclusion of humans in the learning process could be done at different levels depending on who was in control of the process. Active learning requires humans to annotate unlabelled samples, thereby expanding and improving the training dataset. Iterative retraining involves human input to refine the model based on newly labelled data, often using a pool-based scenario for querying instances. Confidence scoring prompts human intervention when predictions fall below a certain threshold, ensuring model accuracy and accountability [263,278,279]. Effective feedback collection, including identifying false positives and false negatives, aids in continuous model refinement [263]. Despite its effectiveness, challenges persist in addressing human time constraints and developing expert systems to handle flagged instances, highlighting ongoing concerns in industry applications.

8.3.3. Challenges in training time, inference, accuracy and continuous monitoring

One of the most critical challenges in deploying DL-based model for industrial defect detection lies in balancing inference speed with training time. While advanced models, such as modified YOLOv8 and DGNNet demonstrate strong detection capabilities—achieving up to 98.36% mAP [187,188,205], they require substantial computational resources, limiting their use in low-latency settings such as edge devices or high-speed production lines. In some domains, such as fabric or steel inspection, inference times between 2–10 s may be acceptable; however, real-time applications demand much faster results. Lightweight models such as MobileNet-based architectures [156] offer faster inference but tend to under-perform on complex or low-contrast defects. Similarly, YOLO variants enhanced with attention or pyramid modules [182,186,203] improves accuracy but often increase latency due to added complexity. While inference time is often acceptable in many applications, training time becomes a critical bottleneck in systems designed for continuous monitoring and model updating. In production settings where new defect types emerge over time, frequent retraining is necessary, raising the question of how to adapt efficiently without compromising the model performance.

To address this, continuous monitoring and human feedback are essential for assessing how well the model performs on new data [276, 277]. HITLA systems allow uncertain or low-confidence predictions to be flagged for manual review, which not only supports quality control but also generates new labelled data for retraining. However, retraining strategies can vary depending on the data; some update the model incrementally with small datasets, while others wait to collect a large dataset before fine-tuning or retraining from scratch. Both approaches face trade-offs: smaller updates risk poor generalisation, while larger ones may suffer from data distribution shifts or increased training time. Moreover, fine-tuning via TL can also lead to catastrophic forgetting or reduced performance in previously well-performing classes for a small dataset [263]. As a result, ongoing research focuses on finding optimal strategies that balance training cost, update frequency, and performance retention. Incorporating efficient learning techniques such as TL, continuous learning, and adaptive fine-tuning is the key to maintaining robust and responsive models in real-world defect detection systems.

8.4. Future industry applications

The complete study reveals that defect detection research is most prevalent in fabric (14–18 papers), steel (12–15 papers), and wood (10–12 papers), driven by industrial relevance and benchmark datasets, such as NEU-DET, DAGM, and TILDA. Moderate research (5–6 papers) exists for metal, ceramic, and glass, while domains such as concrete, plastic, battery, PCB, and engine/gears are less studied (2–3 papers each), often due to limited public datasets or proprietary data restrictions. Critically, only one paper focuses on paper/art paper and three on trading cards, highlighting them as severely underexplored [41,280–282]. These areas lack benchmark datasets, hindering deep learning advancements. Their visual complexity and precision demands make them promising yet challenging frontiers for future research, particularly with targeted dataset creation and augmentation strategies. Currently, the whole process of detecting and classifying defects on trading cards relies on manual and subjective methods, which creates inconsistencies in quality assessment. Therefore, there is a promising chance to explore this field by addressing the challenges of defect detection on trading cards. To assess the quality of trading cards some criteria should be taken into account such as corner defects, edge defects, surface defects and centring measurement [263,283]. These defects are in responsible for assessing the overall quality and condition of the trading cards. Corner defects are irregularities or damages at card corners [263], affecting the size and physical appearance of the card while defects on the edge are imperfections along edges due to wear, or dent. Surface defects include scratches, blemishes and markings, affecting visual appeal. By assessing all these defects one can evaluate trading card quality, aiding collectors. Alternately, widely researched area is defect detection on fabric [40,49,50,156]. Fabric inspection system plays a key role for quality assurance in textile manufacturing. There is an ever-growing demand in the textile factory to substitute the human intensive quality inspection performed with naked eyes by an automated inspection system. This task compelled researchers and practitioners to develop deep CNN-based approaches to isolate fabric defects.

9. Conclusion

Recent advancements in AI and DL techniques have led to successful applications in industrial products for defect detection. Finding a suitable DL model for solving the defect detection problem is very difficult due to the particularities of industrial scenarios and different types of materials. This article provides a comprehensive study of defect detection and classification methodologies by studying more than 150 scientific contributions. The literature review verified that, due to the challenges associated with dataset labelling and annotation, scholars examine unsupervised, self-supervised, semi-supervised and weekly-supervised models in addition to supervised models. Achieving the expected significant detection accuracy and training the model have issues associated with both types of models. Moreover, finding the defective region in a complex pattern poses another type of challenge.

Henceforth, this survey provides a comprehensive review of current research, firstly it describes the use of three approaches (e.g., manual, semi-automated and automated) in defect detection. It is found that the manual defect detection method can assist humans to identify defects but it completely depends on human judgement and needs a prior experimental setup. This problem can be solved by using semi-automated approaches but have some limitations. These methods are not capable of identifying different kinds of defects in different materials. ML and DL have gained popularity in defect detection as they can detect any kind of defect in every kind of material. In many cases, classical ML methods (semi-automated) have shown lower efficiencies than DL methods while the main challenge of DL techniques is depending on large training datasets. Secondly, this article explores the application of DL methods for defect detection on various materials, assessing their accuracy and performance. The DL-based models are

discussed into several aspects: CNN-based, transfer learning and hybrid models for defect classification and encoder-decoder, pyramid networks, Yolo-based model and attention-based models for defect localisation. Our analysis found that encoder-decoder networks are the most prominent techniques in segmentation, while YOLO models with attention mechanisms dominated detection tasks. Additionally, unsupervised, semi-supervised, self-supervised and weakly-supervised models are also cover a good percentage of the overall literature. Furthermore, emerging approaches such as LLMs, foundation models, few-shot learning, and transformer-based methods have also shown increasing attention in recent literature. Finally, discovering the major challenges, it is anticipated that demand for automated inspection will continue to rise in both academic and industry enterprises.

CRedit authorship contribution statement

Lutfun Nahar: Visualization, Software, Investigation, Data curation, Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization. **Mohammad Awrangjeb:** Methodology, Funding acquisition, Conceptualization, Supervision, Formal analysis, Visualization, Writing – review & editing, Validation, Project administration, Investigation. **Md. Saiful Islam:** Validation, Project administration, Investigation, Formal analysis, Visualization, Methodology, Funding acquisition, Conceptualization, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Lutfun Nahar reports financial support was provided by Media8 Pty. Ltd. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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